



CHARUSAT

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Medical Sciences

**Ashok & Rita Patel Institute of Physiotherapy
(ARIP)**

**Master of Physiotherapy(MPT)
(Musculoskeletal Sciences)**

ACADEMIC REGULATIONS & SYLLABUS DETAILS

AY 2019-20



Ashok & Rita Patel
Institute of Physiotherapy

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MEDICAL SCIENCES

ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY

M.P.T. PROGRAMME

COURSE REGULATIONS

1. OBJECTIVES:

The Post Graduate of Physiotherapy programme:

- Integrate knowledge of basic sciences and physiotherapy in order to modify treatment approaches that reflect the breadth and scope of physiotherapy practice.
- Integrate the use of basic principles of research in critical analysis of concepts and findings generated by self and others.
- Practice in an ethical and legal manner that values the professionalism.
- Prepare leaders in the multifaceted roles of clinicians, educators, researchers, and administrators in individual, group, and community contexts model.
- Facilitate commitment to independent thinking and lifelong learning and realization of the intrinsic rewards of these attributes.
- Design a plan of care that synthesizes best available evidence and patient preferences, implementing safe and effective psychomotor interventions, and determining the efficacy of patient outcomes.
- Develops interests for care of the individual and provide physiotherapy services to the community.

2. ELIGIBILITY FOR ADMISSION:

Eligibility of a candidate for admission to Master of Physiotherapy programme will be according to the regulations for admission decided by CHARUSAT norms from time to time.

3. DURATION OF THE COURSE:

The duration of Master of Physiotherapy programme shall be of two academic years (4 semesters).

4. MEDIUM OF INSTRUCTION:

English shall be the medium of instruction for all the courses of study and for the examinations.

5. ATTENDANCE:

A candidate is required to attend at least 80% of the total classes conducted in a semester in each course prescribed for that semester, separately in theory, practical and clinical practice.

6. COURSES OF STUDY:

The courses of M.P.T. programme of two years are given in **ANNEXURE I**.

7. COURSE EVALUATION:

7.1 The performance of every student in each course for university examination will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course;

7.1.2 Final examination by the university through written paper or practical examination or oral examination by the student of combination of any two or more of these, for 70% of the marks of the course.

7.2 Internal evaluation:

7.2.1 The internal assessment is done based on continuous evaluation method. Every semester, there will be two internal examinations for theory and practical. For the award of internal marks in theory and practical, the average of the two tests shall be considered along with other components like attendance, case presentations, microteaching, journal clubs and assignments.

7.3 University (External) Examination:

7.3.1 Every student has to score minimum 40 % of marks in theory and practical examination separately to pass in the final University Examination.

7.4 Aggregate:

7.4.1 Every student has to have an aggregate score of 50 % of marks of both in theory and practical examination separately to pass in the final University Examination and the grade will be awarded based on the aggregate marks.

8. GRADING:

8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Range of Marks (%)	≥80	≥75 <80	≥70 <75	≥65 <70	≥60 <65	≥55 <60	≥50 <55	<50
Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA).

The SGPA and CGPA are defined as follows:

(i) $SGPA = \sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i and $i = 1$ to n , n = number of courses in the semester

(ii) $CGPA = \sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

(iii) No student will be allowed to move further if CGPA is less than 3 at the end of first year.

(iv) In addition to above, the student has to complete the required formalities as per the regulatory bodies.

9. Awards of Degree:

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

9.1.1 He/She should have earned at least minimum required credits as prescribed in course structure,

9.1.2 He/She should have cleared all internal and external evaluation components in every course,

9.1.3 He/She should have secured a minimum CGPA of 5.0 at the end of the programme,

9.1.4 He/She should have completed the dissertation.

9.2 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades in the final semester so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

10. Award of Class:

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction: $10.0 \geq \text{CGPA} \geq 7.5$

First class: $7.5 > \text{CGPA} \geq 6.0$

Second Class: $6.0 > \text{CGPA} \geq 5.0$

Pass class: $5.0 > \text{CGPA} \geq 4.5$

11. Transcript:

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

FACULTY OF MEDICAL SCIENCES

ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY

CHOICE BASED CREDIT SYSTEM

FOR

**MASTER OF PHYSIOTHERAPY
(MUSCULOSKELETAL SCIENCES)**

CHOICE BASED CREDIT SYSTEM

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 3 types of courses: Foundation courses, Core courses, Elective courses.

1.1. Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

1.2. Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme. Environmental science will be a compulsory University core for all Undergraduate Programmes.

B. Programme Core courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

1.3. Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialised or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidate's proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes. As a general guideline, Programme should incorporate 2 University Electives of 2 credits each (total 4 credits).

B. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

1.4. Medium of Instruction

The Medium of Instruction will be English.

Ashok & Rita Patel Institute of Physiotherapy

Vision

To become a leading institute in evidence based education and research by preparing physiotherapists having proficiency and ethical values

Mission

To create physiotherapists with contemporary knowledge, entrepreneurial skills and ethical values to serve society

Program Outcomes: Physiotherapy

PO1 Communication: Demonstrate effective communication and interpersonal skills, which are adapted to meet the needs of diverse individuals and groups

PO2 Diagnosis and plan of care: Develop physiotherapy diagnoses and an individualized plan of care for the management and prevention of movement dysfunction

PO3 Evidence based practice: Apply principles of critical thinking and clinical reasoning to evidence-based physiotherapy practice.

PO4 Teaching and learning principles: Apply learning and teaching principles in educational, clinical and community settings

PO5 Team Member: Participate as an effective intra- and inter-professional team member

PO6 Ethical and legal standards: Adhere to safe, ethical and legal standards of current practice (as identified by professional organizations, central and state law and accrediting bodies)

PO7 Professional Responsibility and commitment: Develop responsibility and commitment to the profession and society through life-long learning and involvement in activities beyond job responsibilities

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSITITUTE OF PHYSIOTHERAPY
MASTER OF PHYSIOTHERAPY (M.P.T.)
BRANCH: MUSCULOSKELETAL SCIENCES
1ST SEMESTER

Course Code	Course Title	Teaching Scheme				Examination Scheme				
		Contact Hrs.			Credit	Theory		Practical		Total
		Theory	Practical	Total		Internal	External	Internal	External	
HS703.01F	Foreign Language (French/ German)		2	2	2	-	-	30	70	100
HS704F	Academic Speaking									
PT795.01	Health & Physical Activity	-	2	2	2	-	-	30	70	100
PT751	Clinical Education I	-	16	16	8	-	-	60	140	200
PT793	Research Methodology & Biostatistics	4	2	6	5	30	70	30	70	200
PT794	Measurement in Physiotherapy	4	-	4	4	30	70	-	-	100
PT966	Physiotherapy Education									
PT873	Research Ethics	3	2	5	4	30	70	30	70	200
PT875	Imaging in Physiotherapy									
TOTAL		11	24	35	25	90	210	180	420	900

2ND SEMESTER

Course Code	Course Title	Teaching Scheme				Examination Scheme				
		Contact Hrs.			Credit	Theory		Practical		Total
		Theory	Practical	Total		Internal	External	Internal	External	
HS705 F	Academic Writing	-	2	2	2	-	-	30	70	100
PT796.01	Fitness & Nutrition	-	2	2	2	-	-	30	70	100
PT754	Clinical Education II	-	16	16	8	-	-	60	140	200
PT797	Basic Sciences (Biomechanics, Pathomechanics & Ergonomics)	4	-	4	4	30	70	-	-	100
PT798	Exercise Physiology & Nutrition	5	-	5	5	30	70	-	-	100
PT868	Exercise Programming									
PT963.01	Literature Review	3	2	5	4	30	70	30	70	200
PT870	Neonatal Care									
TOTAL		12	22	34	25	90	210	150	350	800

3RD SEMESTER

Course Code	Course Title	Teaching Scheme				Examination Scheme				
		Contact Hrs.			Credit	Theory		Practical		Total
		Theory	Practical	Total		Internal	External	Internal	External	
PT901.01	Clinical Education III	-	10	10	5	-	-	30	70	100
PT 902.02	Dissertation I	-	15	15	11	-	-	120	280	400
PT 903.01	Physical & Functional Diagnosis for Musculoskeletal Sciences	4	2	6	5	30	70	30	70	200
PT 904.01	Advanced Physiotherapeutics for Musculoskeletal Sciences	3	2	5	4	30	70	30	70	200
TOTAL		7	29	36	25	60	140	210	490	900

4TH SEMESTER

Course Code	Course Title	Teaching Scheme				Examination Scheme				
		Contact Hrs.			Credit	Theory		Practical		Total
		Theory	Practical	Total		Internal	External	Internal	External	
PT905.01	Clinical Education IV	-	10	10	5	-	-	30	70	100
PT906.01	Dissertation II	-	26	26	20	-	-	200	400	600
TOTAL		-	36	36	25	-	-	230	470	700

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS703.01 F LANGUAGES (French)

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
I	HS703.01	LANGUAGE S (French)	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	Introduction to French Language Facts and figures about French Language Basic French Linguistics * Alphabets * Accents * Liaison * Nasalization French Culture, Differ between French and English Grammar : • Subject Pronoun • Verbs : (être ,avoir, habiter,regarder,manger ... “er” verb) • Form of address • Numbers (1 to 20) • Nouns and plurals of nouns • The expression : C’est ,Il y a Presentation: 1) Self Introduction 2) Question and answering Dialogue	08
2	Grammar : • Definite articles • Indefinite articles • Present tense ○ Positive Forms ○ Negative Forms • Numbers (21 to 100 , 100-1000) • Days ,Months ,Family • Verbs: (aller,venir,finir,pouvoir,vouloir “ir “ verb) Social Links:	08

	1) My family & relations 2) Appointments 3) Gathering information from someone Dialogue	
3	Grammar : • Common Adjectives • Comparative Adjectives • Common Adverbs • Interrogative Forms • The expression : “On” • Directions , Countries, Nationalities, Seasons, Weather, • Professions • Verbs: (Prendre, Apprendre, Comprendre, faire ... “re “ verb) Work , Study and Travel 1) Job/ Profession 2) Ticket Reservation (At Bus/At Railway/At Airport) Dialogue	08
4	Grammar : 1) Common Prepositions 2) Common Conjunctions 3) Past Tense 4) Future Tense 5) Colors ,Shapes, Animals ,Vegetables, Fruits 6) Verbs: (“er”, “ir”, ”re” etc...) Food & Shopping 1) Buy a vegetables and fruits 2) Any Conversation between Customer and Vendor (At Mall/ At Restaurant / At Market) Dialogue	06
Total		30

III. Instruction Methods and Pedagogy

The course is based on pragmatic learning. Teaching will be facilitated by Reading Material, Discussion, Task-based learning, assignments and various interpersonal activities like group/pair work, independent and collaborative work, presentations, etc.

IV. Evaluation:

The students will be evaluated continuously in the form of their consistent performance. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

The students’ performance in the course will be evaluated on a continuous basis through the following components:

Internal Evaluation:

Sl. No.	Component	Number	Marks per incident	Total Marks
1	Assignment / Presentation/ Task	5	5	25
2	Attendance and Class Participation			05
Total				30

External Evaluation

The University Practical Examination will be for 70 marks and will test the LSRW skills in the French Language.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Term Work, Viva and Practical (LSRW)	-	70	70
Total				70

Course Outcome (COs):

After completion of the course, the student would:

CO1	Have a basic overview of the French language and culture.
CO2	Develop a basic understanding of the usage of French language such as alphabets,numbers,months , colour etc.
CO3	Learn the basics of French Grammar and their usages
CO4	Learn to use basic language functions such as introducing oneself, answering basic Wh questions
CO5	be able to initiate conversations/dialogues in the French language confidently.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	-	-	-	-	-	2
CO2	3	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-
CO5	3	-	-	-	2	-	2

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Note: The reference/ reading material will be provided in consultation with the expert.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS704 F: ACADEMIC SPEAKING (Sem-I)

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
1	HS704 A/B/C/D/E/F/G/ H	Academic Speaking	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	Foundations of Advance Communication - Meaning and Definition of Advance Communication - Advance Communication in Digital, Social, Mobile World - Strategies for Advance Communication - Meaning and Concept of Academic Language - High Frequency Academic Vocabulary	04
2	Art of Conversation - Describing people, places and things - Expressing opinions - Making suggesting - Persuading someone - Interpreting and Summarizing	06
3	Science of Power Speaking - Phonemes - Word Stress - Pronunciation - Intonation - Pause - Register - Fluency - Prosody	06

	• <i>Lexical Range</i>	
4	<i>Academic Speaking Application – Part I</i> • <i>Art of Oratory</i> • <i>Formal Presentation</i> • <i>Speech Analysis – Decoding Best Speeches</i>	08
5	<i>Academic Speaking Application – Part II</i> • <i>Job Interview</i> • <i>Group Discussion</i> • <i>Meeting</i>	06
Total		30

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies, group work, independent and collaborative research, presentations etc.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	I-Talk	1	10	25
2	Situational Speaking	1	05	
3	Case Study - Speech Analysis	2	10	
4	Attendance and Class Participation	-		05
Total				30

External Evaluation

The University Practical Examination will be for 70 marks and will test the advance communication skills and academic speaking.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

Course Outcome (COs):

After completion of the course the student would :

CO1	understand and demonstrate advance communication skills and academic speaking.
CO2	demonstrate linguistic competence
CO3	demonstrate performing ability at group discussion and personal interview.
CO4	demonstrate the formal presentation skills.
CO5	demonstrate ability to communicate in diverse situations

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-
CO3	3	-	-	-	3	-	2
CO4	3	-	-	-	-	-	-
CO5	3	-	-	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

V. Reference Books

- Headway Academic Skills - Level 1: Listening, Speaking and Study Skills Student's Book Paperback

VI. Reading

- **Unit 1:** Business communication Today (Thirteenth Edition) by Courtland L. Bovee, John V. Thill and Roshan Lal Raina
- **Unit 2:** Effective Speaking Skills by Terry O' Brien
- **Unit 2:** Speak Better Write Better by Norman Lewis
- **Unit 2:** Well Spoken: Teaching Speaking to All Students by Erik Palmer
- **Unit 3:** Let Us Hear Them Speak : Developing Speaking – Listening Skills in English by Jayshree Mohanraj (Publisher – Sage Publication)
- **Unit 4:** The craft of scientific presentations: Critical steps to succeed and critical errors to avoid. New York: Springer by Michael Alley
- **Unit 4:** Presentation Skills in English by Bob Dignen (Publisher: Orient Black Swan)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester I
PT795.01 HEALTH & PHYSICAL ACTIVITY

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
-	2	2	-	2	2	-	100	100

A. Outline of The Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	INTRODUCTION TO HEALTH	10
2.	PHYSICAL ACTIVITY	15
3.	INTRODUCTION TO YOGA	15

Total Hours (Theory): 40

Total Hours (Practical): 00

Total Hours: 40

B. Content Of The Course:

1	INTRODUCTION TO HEALTH	10 hrs	25%
1.1	What is health?		
1.2	Healthcare delivery system: Developing Countries, Developed Countries		
1.3	Human anatomy, physiology & physical fitness		
1.4	Basics of Nutrition		
1.5	Life style disorders – obesity & diabetes		
2	PHYSICAL ACTIVITY	15 hrs	37.5%
2.1	What are Physical activity, exercise, physical fitness and epidemiology?		
2.2	Measurement of Physical Activity in individuals		
2.3	Physical Activity – Theoretical Perspective: Self-determination, trans theoretical		
2.4	Physical Activity and mental health – Body image, depression, problem with exercise		
2.5	Barriers & Facilitators of Physical Activity		
3	INTRODUCTION TO YOGA	15 hrs	37.5%
3.1	What is yoga?		
3.2	Types of yoga		
3.3	Benefits of yoga to various body systems		
3.4	Asanas, Pranayam		
3.5	Yoga therapy for various back pain, asthma, stress, hypertension, diabetes		

Course outcomes (COs)

At the end of the course, the student will be able:

CO1	Describe interrelation between physical activity, its physiological effects and changes in human anatomy.
CO2	Understand the health-related benefits of physical activity and risks associated with physical inactivity
CO3	Realize the areas of nutrition, cardiovascular health, diseases related to physical activity, stress management, substance use and abuse, and sexually transmitted diseases
CO4	Describe correlation of physical activity and mental health
CO5	Describe yoga, different types yoga and its application in different health disorders

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	-	3	3	-	-	2
CO2	-	3	3	2	-	2	3
CO3	-	2	2	3	-	2	2
CO4	1	2	2	2	-	1	2
CO5	3	3	1	3	3	-	3

Recommended Study Material:

❖ Textbooks:

1. ACSM's "Health Related Physical Fitness Assessment Manual Lippincott Williams and Walkins USA, 2005.
2. Nilima Patel (2008) Yoga and Rehabilitation, Jaypee Publication, India

❖ Reference Books:

1. Biddle, S. J. H., & Mutrie, N. (2008). Psychology of physical activity. London: Routledge
2. B.C. Rai. Health Education and Hygiene Published by Prakashan Kendra

3. Puri. K.Chandra.S.S. (2005). Health and Physical Education. New Delhi: Surjeet Publications

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester I
PT751 - CLINICAL EDUCATION I

Credit Hours

Teaching Scheme				Examination Scheme				
Contact Hrs.			Credit	Theory		Practical		Total
Theory	Practical	Total		Internal	External	Internal	External	
-	16	16	8	-	-	60	140	200

Total Hours: 320 Hours

A. Outline of the Course:

Sr. No.	Skill Sets
1	Display professional and ethical behavior while in the assigned treatment environment.
2	Demonstrate effective and appropriate oral, nonverbal and written communication skills.
3	Assure safety of patient and self throughout patient care.
4	Assure patient privacy, rights, and dignity.
5	Examination • Muscle Strength • Muscle length • Special test to identify various conditions • Coordination • Posture • Balance / fall risk assessment • Gait • Activities of daily living • Pain • Environmental assessment • Differential diagnosis for various conditions
6	Provide competent provision of physiotherapy interventions, including: Therapeutic exercise A. Aerobic Capacity/Endurance Conditioning or Reconditioning

	B. Balance, coordination, and agility training C. Body mechanics and postural stabilization D. Flexibility exercises E. Gait and locomotion training F. Relaxation G. Strength, power, and endurance training for head, neck, limb, trunk, and ventilatory muscles
7	Functional training in self-care and home management A. Activities of daily living (ADL) training B. Device and equipment use and training C. Injury Prevention or reduction
8	Manual therapy techniques
9	Use clinical problem-solving skills in patient care.
10	Anthropometric characteristics
11	Professional Commitment and Attitude Towards improving the quality of care

Clinical training comprises all of the formal and practical "real-life" learning experiences provided for students to apply knowledge and skills in the clinical environment. Experiences would include those of short and long duration (e.g., part-time, full-time) and those that provide a variety of learning experiences (e.g., rotations on different units within the same practice setting such as Musculoskeletal Physiotherapy OPD & IPD etc.) to include comprehensive care of patients. Each student will be under the supervision of a clinical supervisor at the clinical education site who directly instructs and supervises students during their clinical learning experiences. Clinical supervisors will evaluate each student twice during clinical training (mid-term and final) using a standard evaluation tool.

The student will be formally evaluated through **Objective Structured Clinical Examination (OSCE)** by the internal and external examiners.

B. Course Outcomes:

At the end of the course, the students will be able to

CO1	Communicate effectively with patients, family/carers, multidisciplinary team and colleagues
CO2	Demonstrate skills in history taking, complete review of systems and physical examination.
CO3	Identify appropriate assessment tools for specific musculoskeletal conditions & conduct systematic assessments

CO4	Develop a patient centered evidence-based management plan.
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Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	-	-	-	2
CO2	-	3	2	-	-	3	-
CO3	-	2	3	-	-	-	-
CO4	-	2	2	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY FACULTY
OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester I
PT793 RESEARCH METHODOLOGY & BIOSTATISTICS**

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
4	2	6	4	1	5	100	100	200

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	INTRODUCTION TO RESEARCH	05
2.	STUDY DESIGNS	10
3.	SAMPLING	10
4.	DATA COLLECTION	15
5.	BIOSTATISTICS	35
6.	PUBLISHING AND PRESENTING RESEARCH	05

Total hours (Theory): 80 Hrs.

Total hours (Practical): 40 Hrs.

Total hours: 120 Hrs.

A. Detailed Syllabus:

1	INTRODUCTION TO RESEARCH	05 Hours	6%
1.1	Definition and need for research		
1.2	History of research in physiotherapy		
1.3	Research problem and research question		
1.4	Variables and its types		
1.5	Literature search using electronic databases		
2	STUDY DESIGNS	10 Hours	13%
2.1	Bias in research: types, the ways in which it can threaten study validity, measures to minimize bias in research		
2.2	Quantitative and qualitative research designs		
2.3	Internal and external validity of research		
2.4	Commonly used quantitative research designs and their major strength and limitations: Descriptive, observational, analytical and experimental study designs		
2.5	Overview of qualitative research methods		
3	SAMPLING	10 Hours	12%
3.1	Populations and samples		
3.2	Sampling methods: Probability and non-probability sampling methods		
3.3	Assignment to groups		
3.4	Sample size calculation		
4	DATA COLLECTION	15 Hours	19%
4.1	Primary versus secondary data collection		
4.2	Measurement error and the factors contribute to measurement error		
4.3	Outcome measures: types, validity, reliability and its uses in research		
5	BIOSTATISTICS	35 Hours	44%

5.1	Data display and summary		
5.2	Summary statistics for quantitative data		
5.3	Statements of probability and confidence intervals		
5.4	Choosing appropriate statistics		
5.5	Interpreting results of statistics such as p-value, confidence interval		
5.6	P-values, power, type I and type II errors		
5.7	Difference between clinical and statistical significance		
5.8	Use of computer software: Excel, SPSS		
6	PUBLISHING AND PRESENTING RESEARCH	05 Hours	6%
6.1	Types of publication		
6.2	Peer review process		
6.3	Components of research manuscript: IMRAD format		
6.4	Introduction to the EQUATOR network reporting guidelines for various study design		
6.5	Guidelines for presenting research: platform presentation, poster presentation		

Course Outcomes (COs):

At the end of the course, the student will be able to

CO1	Develop understanding on various kinds of research, objectives of doing research, research process, research designs and sampling techniques.
CO2	Able to distinguish a purpose statement, a research question or hypothesis, and a research objective.
CO3	Identify and discuss the role and importance of research in the physiotherapy.
CO4	Differentiate between qualitative and quantitative research.
CO5	Have basic knowledge of data analysis-and hypothesis testing procedures
CO6	Prepare a research proposal

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	2	2	1	-	3	3
CO2	-	1	-	1	-	2	1
CO3	1	-	-	2	1	-	-
CO4	-	-	1	3	1	3	2
CO5	-	-	-	-	-	3	3
CO6	-	-	2	1	3	3	1

Recommended Study Material:

❖ Textbooks:

1. Rehabilitation research: Principles and applications. Elizabeth Domholdt (2005). 3rd ed., 592 pp., Elsevier Saunders, St Louis, MI, USA.
2. Statistics at Square One. Michael J. Campbell, T. D. V. Swinscow. October 2009, ©2010, BMJ Books

❖ Reference Books:

1. Research in Physical Therapy. Editor-Christopher E. Bork Author. Published by J.B. Lippincott Co. (1993).
2. Greenhalgh T. How to read a paper: The basics of evidence-based medicine. John Wiley & Sons; 2014 Feb 26.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester I
PT794 MEASUREMENT IN PHYSIOTHERAPY

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
4	-	4	4	-	4	100	-	100

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	MEASUREMENT IN CLINICAL PRACTICE	12
2.	VALIDITY	23
3	MEASUREMENT ERROR	15
4	COMMON MEASUREMENT TOOLS USED IN PHYSIOTHERAPY	30

Total hours (Theory): 80

Total hours (Practical): 00

Total hours: 80

B. Detailed Syllabus:

1	MEASUREMENT IN CLINICAL PRACTICE	12 Hours	15%
1.1	Need for standardized measurement in clinical practice		
1.2	Purpose of measurement in clinical practice: Screening, diagnosis, prognosis and treatment outcomes		
1.3	Levels of measurement: Nominal, ordinal, interval and ratio level		
1.4	WHO's ICF as a framework for measuring functioning and disability Classification of measurement tools: physiologic measures and biomarkers, performance based measures and self-reports, norm referenced and criterion referenced tests, disease specific and generic measures		
2	VALIDITY	23 Hours	29%
2.1	Content validity		
2.2	Construct Validity including convergent, discriminant concurrent and criterion validity		
2.3	Responsiveness to change		
2.4	Methods to evaluate various types of validity of measurement tools including responsiveness to change (MCID)		
2.5	Interpretation of validity assessment		
3	MEASUREMENT ERROR	15 Hours	19%
3.1	Measurement error: definition and sources of error		
3.2	Strategies to eliminate measurement error		
3.3	Estimation of measurement error associated with a measurement tool		
3.4	Reliability (test-retest, inter-intra rater reliability)		
3.5	Interpreting measurement error: Standard error of measurement (SEM) & The Bland- Altman plot , minimum detectable change (MDC), Receiver Operating Characteristic (ROC) curve		
4	COMMON MEASUREMENT TOOLS USED IN PHYSIOTHERAPY	30 Hours	38%
	Validity, interpretability and clinical utility of measures of the following:		
4.1	Measures of muscle performance: strength, tone, flexibility		
4.2	Measures of joint functions		
4.3	Measures of functional status		
4.4	Measures of pain		
4.5	Measures of balance and gait		
4.6	Measures of cardiovascular functions		
4.7	Screening tests of cognitive functions, nerve, muscle and other soft tissue integrity		

Course Outcomes (COs):

At the end of the course the student will be able to:

CO1	Recognize the value and importance of using standardized measurement tools in routine clinical practice.
CO2	Communicate the problems and biases that may arise when assessing constructs in clinical practice
CO3	Identify and define different types of reliability, distinguishing among types of reliability and their unique insights into the assessment of outcomes
CO4	Define measurement error, its types and recognize its influence in interpreting measurement scores
CO5	Evaluate and interpret the validity and errors of measurement tools in making clinical decisions about patient care.

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	-	3	-	-	-	-
CO2	-	2	3	-	-	-	-
CO3	-	-	3	2	-	-	-
CO4	-	-	3	-	-	2	-
CO5	-	2	3	2	-	2	-

Recommended Study Materials:

❖ Textbooks

1. Rehabilitation research: Principles and applications. Elizabeth Domholdt (2005). 3rd ed., 592 pp., Elsevier Saunders, St Louis, MI, USA.
2. Finch E, Brooks D, Stratford P, Mayo N. Physical Rehabilitation Outcome Measures: A Guide to Enhanced Clinical Decision Making. Lippincott Williams and Wilkins; 2nd Revised edition

❖ Reference Books

1. Portney LG, Watkins MP, Foundations of clinical Research: Applications to practice. FA Davis & Company; 2015.
2. Mokkink LB, Terwee CB, Patrick DL, Alonso J, Stratford PW, Knol DL, Bouter LM, De Vet HC. The COSMIN checklist for assessing the methodological quality of studies on measurement properties of health status measurement instruments: an international

Delphi study. Quality of life research. 2010 May 1;19(4):539-49.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY

MPT (Musculoskeletal Sciences) - Semester I
PT966 PHYSIOTHERAPY EDUCATION

Credits Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
3	2	5	3	1	4	100	100	200

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	TEACHING & LEARNING	10
2.	CURRICULUM DESIGNING	10
3.	PRINCIPLES & METHODS OF TEACHING	10
4.	MEASUREMENT AND EVALUATION	10
5.	GUIDANCE AND COUNSELING AND AWARENESS PROGRAMME	10
6.	PHYSIOTHERAPY PROFESSION AND STAFF ROLES	10

Total hours (Theory): 60 Hrs.

Total hours (Practical): 40 Hrs.

Total hours: 100 Hrs.

Detailed Syllabus:

1	CONCEPT OF TEACHING AND LEARNING	10 Hours	10%
1.1	Meaning and scope of Educational Psychology.		
1.2	Meaning and Relationship between teaching and learning.		
1.3	Learning Theories.		
1.4	Dynamics of behavior.		
1.5	Individual Meaning and concept.		
2	CURRICULUM	10 Hours	10%
2.1	Basis of curriculum formulation.		
2.2	Framing objectives for curriculum.		
2.3	Process of curriculum development and factors involved.		
2.4	Evaluation of curriculum differences.		
2.5	Curriculum planning – Integrated teaching, Problem based learning, Evidence based medicine.		
2.6	Skill development- Clinical skills, Communication skills, Counseling skills.		
3	PRINCIPLES AND METHODS OF TEACHING	10 Hours	10%
3.1	Bloom's taxonomy of instructional objectives.		
3.2	Writing instructional objectives in behavioral terms.		
3.3	Unit planning, Lesson planning.		
3.4	Lecture, Demonstration Discussion, Seminar, Assignment.		
3.5	Types of teaching aids.		
3.6	Principles of selection, preparation and use of audio-visual aides.		
4	MEASUREMENT AND EVALUATION	10 Hours	10%
4.1	Nature of educational measurement: meaning, process, types of tests.		
4.2	Construction of an achievement test and its analysis.		
4.3	Standardized test.		
4.4	Introduction of some standardized tools, important tests of intelligence.		
4.5	Aptitude and personality.		
4.6	Continuous and comprehensive evaluation.		
5	GUIDENCE AND COUSSELLING AWARENESS PROGRAMME	10 Hours	10%
5.1	Meaning & concepts of guidance and counseling.		
5.2	Principles of guidance and counseling.		
5.3	Awareness and guidance to the common people about health and diseases.		

6	PHYSIOTHERAPY PROFESSION AND STAFF ROLES	10 Hours	10%
6.1	Physiotherapy: Definition and Development.		
6.2	Physiotherapy practice in India and their demands. Physiotherapy services in rural and urban areas.		
6.3	History taking, assessment, tests, Patient communication, documentation of findings, treatment organization and planning/execution for intervention.		
6.4	Documentation of rehabilitation assessment and management using International		
6.5	Standardized tests and scales used in various types of cases for assessment and interpretation in Physiotherapy practice.		
6.6	Roles of Physiotherapy Director, Physiotherapy Supervisor, Physiotherapy Assistant, Physiotherapy, Occupational therapist, Home Health Aide and Volunteer		

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Interpret the concept of teaching & learning
CO2	Design a curriculum for basic and advanced courses
CO3	Deliver the class in an effective manner by engaging all
CO4	Apply the managerial skills and knowledge in physiotherapy profession
CO5	Justify various teaching learning tools for effective assessment practices
CO6	Create counselling and awareness programme for various groups of patients

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	-	2	2	-	-	-
CO2	1	-	1	1	-	-	-
CO3	1	2		-	2	-	-
CO4	2		-	1	-	1	2
CO5	-	1	3	2	1	-	-
CO6	1	-	-	-	2	2	3

Recommended Study Material:

❖ Textbooks:

1. Developing a Pedagogy of Teacher education: Understanding teaching and learning about teaching.

❖ Reference Books:

1. Physical Therapy Administration & Management by Hickik Robert J

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK AND RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester I
PT873 RESEARCH ETHICS

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
4	-	4	4	-	4	100	-	100

A.OUTLINE OF THE COURSE:

Sr. No.	Title of the unit	Minimum number of hours
1.	HISTORICAL PERSPECTIVES AND GUIDELINES TO HUMAN RESEARCH ETHICS	03
2.	NATIONAL & GLOBAL GUIDELINES, CODES AND REGULATIONS	15
3.	RESPECT FOR PERSONS: INFORMED CONSENT AND ASSENT PROCESS, VULNERABLE POPULATIONS	15
4.	SCIENTIFIC INTEGRITY & DATA MANAGEMENT	10
5.	PUBLICATION, PEER REVIEW AND AUTHORSHIP	10
6.	CONFLICT OF INTEREST & COMMITMENT	07

Total hours (Theory): 60 Hrs.

Total Hours (Practical):00 Hrs.

Total Hours (Practical): 00 Hrs.

B.Detailed Syllabus:

1.	HISTORY AND FRAMEWORK FOR ETHICAL CLINICAL RESEARCH	03 Hours	5 %
1.1	Ethics of clinical research, Historical Perspectives on Healthcare Research using human participants		
2.	NATIONAL & GLOBAL GUIDELINES, CODES AND REGULATIONS	15 Hours	25 %
2.1	Belmonte report, Nuremberg Code of Medical ethics, Helsinki Declaration CIOMS/WHO guidelines for epidemiological studies		
2.2	ICMR Guidelines for Biomedical Research using human participants: Ethical Principles of Ethical Research viz. essentiality, voluntariness, informed consent and community agreement, non-exploitation, privacy and confidentiality, precaution and risk minimization, Accountability and transparency professional conduct, public domain, totality of responsibility		
3.	RESPECT FOR PERSONS: INFORMED CONSENT AND ASSENT PROCESS, VULNERABLE POPULATIONS	15 Hours	25 %
3.1	Informed Consent Process, documentation of informed consent, waivers of informed consent, Assent and Permission for Children's Participation in research, compensation issues for research participation		
3.2	Defining and determining vulnerability in context of health care research, Beneficence & Justice, Vulnerability and cognitive impairment, Inclusion of non-competent subjects, Safeguards for vulnerable and non-competent subjects,		
3.3	Ethical Concerns for Vulnerable Population involving pregnant females, human fetuses, neonates, children, persons who are physically handicapped, mentally disabled, economically disadvantaged, educationally disadvantaged, Institutionalized		
4.	SCIENTIFIC INTEGRITY & DATA MANAGEMENT	10 Hours	16.67 %
4.1	Importance of scientific integrity in healthcare research, behaviors that affect scientific integrity, allegations of scientific misconduct, adverse events, scientific record, maintenance, responsibility of researchers for maintaining accuracy of scientific records, data retention		
4.2	Institutional Ethics Committee : Mandate, Role & composition of IEC in ensuring scientific integrity, Proposal Review Process and decision making		
5.	PUBLICATION, PEER REVIEW AND AUTHORSHIP	10 Hours	16.67 %
5.1	Purpose of publication, replication of results of new findings into clinical practice, avoiding fragmentary publication		

5.2	Authorship Issues: ICMJE guidelines to authorship, roles and responsibilities of authors and contributors		
5.3	Peer review: Purpose of peer review, elements of peer review		
6.	CONFLICT OF INTEREST & COMMITMENT	07 Hours	11.66 %
6.1	Defining COI for human research participants including authors, contributors, editorial boardmembers, reporting COI		

Course Outcomes (COs):

CO1	Describe the history and importance ethics of human subjects protections
CO2	Identify research activities that involve human subjects ethics
CO3	Discover the risks a research project might pose to participants and understand how to minimize the risks posed by a research project
CO4	Describe additional protections needed for vulnerable populations
CO5	Describe appropriate procedures for recruiting research participants and obtaining informed consent
CO6	Identify the different professional committees that monitor human subjects protections

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	3	-	-	-
CO2	-	3	-	-	-	3	-
CO3	-	-	-	3	3	3	-
CO4	3	-	-	-	3	-	-
CO5	-	-	3	3	-	-	-
CO6	-	-	-	-	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation

Recommended Study Material:

❖ Textbooks

1. Emanuel, E. J., Grady, C. C., Crouch, R. A., Lie, R. K., Miller, F. G., & Wendler, D. D. (Eds.). (2011). The Oxford textbook of clinical research ethics. Oxford University Press.

❖ Scientific Papers

1. Walanj, A. S. (2014). Research ethics committees: need for harmonization at the national level, the global and Indian perspective. Perspectives in clinical research, 5(2), 66.
2. Sanmukhani, J., & Tripathi, C. B. (2011). Ethics in clinical research: The Indian perspective. Indian journal of pharmaceutical sciences, 73(2), 125.

3. Bhutta, Z. A. (2002). Ethics in international health research: a perspective from the developing world. *Bulletin of the World Health Organization*, 80(2), 114-120.

❖ **Reference Books**

1. Long, T., & Johnson, M. (2007). *Research ethics in the real world: issues and solutions for health and social care*. Elsevier Health Sciences.
2. McNamee, M. J., Olivier, S., & Wainwright, P. (2006). *Research ethics in exercise, health and sports sciences*. Routledge.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MEDICAL SCIENCES

ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY

MPT (Musculoskeletal Sciences) - Semester I

PT875 IMAGING IN PHYSIOTHERAPY

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
3	2	5	3	1	4	100	100	200

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	INTRODUCTION TO IMAGING	12
2.	RADIOLOGIC EVALUATION, SEARCH PATTERNS AND DIAGNOSIS	8
3.	RADIOLOGIC EVALUATION OF MUSCULOSKELETAL DISORDERS	16
4.	RADIOLOGIC EVALUATION OF NEUROLOGICAL DISORDERS	12
5.	RADIOLOGIC EVALUATION OF CARDIORESPIRATORY DISORDERS	12

Total hours (Theory): 60 Hrs.

Total hours (Practical): 40 Hrs.

Total hours: 100 Hrs.

B. Detailed Syllabus:

1	INTRODUCTION TO IMAGING	12 Hours	20%
1.1	Introduction to: Radiology (X-Ray), Computer tomography (CT), Magnetic resonance imaging (MRI), Diagnostic ultrasound, Angiography, Doppler and Echocardiogram.		
1.2	Role of the various imaging modalities: strengths, limitations, relative cost, associated risks, ionizing radiations, Contrast media-related issues.		
1.3	Integration of Imaging Into Physiotherapy Practice.		
2	RADIOLOGIC EVALUATION, SEARCH PATTERNS AND DIAGNOSIS	8 Hours	13.3%
2.1	Search pattern: the ABCs of radiologic analysis.		
2.2	Understanding the image and the radiologic report.		
3	RADIOLOGIC EVALUATION OF MUSCULOSKELETAL DISORDERS	16 Hours	26.6%
3.1	Routine radiologic evaluation & Advance imaging evaluation of common spine, upper limb and lower limb disorders.		
4	RADIOLOGIC EVALUATION OF NEUROLOGICAL DISORDERS	12 Hours	20%
4.1	Routine radiologic evaluation & Advance imaging evaluation of common neurological disorders.		
5	RADIOLOGIC EVALUATION OF CARDIORESPIRATORY DISORDERS	12 Hours	20%
5.1	Routine radiologic evaluation & Advance imaging evaluation of common cardiorespiratory disorders.		

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Understand the concept of imaging techniques used in musculoskeletal, neurological and cardiorespiratory disorders.
CO2	Interpret the imaging based on the evidence, ethical and legal standards.
CO3	Integrate imaging concept in clinical assessment and management
CO4	Practice effective inter-professional communication in rehabilitation team.
CO5	Criticize the imaging study and communicate effectively with various stakeholders.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	3	3	3	-	-	-
CO2	-	-	3	-	-	3	-
CO3	-	3	3	-	-	-	-
CO4	-	-	-	-	3	-	-
CO5	3	-	-	3	-	-	-

Recommended Study Material:**❖ Textbooks:**

1. Lynn N. Mc Kinnis, Fundamentals of Musculoskeletal Imaging, 4th Edition, Philadelphia, F.A. Davis, 2014.
2. Lawrence Bost, Suhny Abbara, Cardiac Imaging: The Requisites, 4th edition, Elsevier.
3. Jeffrey P Kanne, Clinically Oriented Pulmonary Imaging (Respiratory Medicine).
4. Pina Sanelli, Pamela Schaefer, Laurie Loevner, Neuroimaging: The Essentials (Essentials Series), First Edition.

❖ Reference Books:

1. Lynn N McKinnis, Michael Mulligan, Musculoskeletal Imaging Handbook: A Guide for Primary Practitioners. Philadelphia, F.A. Davis, 2014.
2. Ramón Ribes, Joan C Vilanova, Learning Musculoskeletal Imaging. Springer Heidelberg; London, New York, 2010.
3. Marcelo F. Di Carli, Raymond Y. Kwong, Novel Techniques for Imaging the Heart: Cardiac MR and CT (American heart association).
4. Robert L Kilkins, James R Dexter, Albert J Heuer, Clinical assessment in respiratory care, 6th Edition by Mosby, Elsevier.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS705 (F): ACADEMIC WRITING

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
II	HS705	Academic Writing	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title / Topic	Classroom Contact Hours
1	Academic Writing and Research Process • Introduction to Academic Writing • Academic Writing as a Part of Research • Types of Academic Writing • Features of Academic Writing • Importance of Good Academic Writing in various Academic Works	05
2	Anatomy of Academic Writing • Academic Vocabulary • Simple and Complex Sentences • Organizing Paragraphs • The Writing Process • Adopting Academic Writing Style	05
3	Key Academic Skills • Note – taking • Note – making • Paraphrasing • Summarizing	05
4	Accuracy in Academic Writing • Lexical Range • Academic Language and Structures • Elements of Writing • <i>Proof Reading, Editing, and Rewriting</i>	05
5	Using and Citing Sources of Ideas • Academic Texts and their Types • Intellectual Honesty in Academic Writing • Avoiding Plagiarism – Idea Theft • Degrees of Plagiarism • Types of Borrowing • Anatomy of Citations • <i>Common Citation Styles</i>	05

6	Contemporary Practices in Academic Writing - Analytical Essays - Graph / Table / Process Interpretation and Description - Writing Reports - Writing Research / Concept Papers	05
Total		30

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like writing, group work, independent and collaborative research, etc.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Paragraph Writing	1	3	03
2	Note-taking / Note-making	1	3	03
3	Paraphrasing / Summarizing	1	4	04
4	Essay Writing	1	5	05
5	Concept Paper Writing	1	10	10
5	Attendance and Class Participation			05
Total				30

External Evaluation

The University Practical Examination will be for 70 marks and will test the professional communication skills and academic writing skills of the students.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical /Quiz/ Project / Academic Writing	-	70	70
Total				70

Course Outcome (COs):

After completion of the course, the student would:

CO1	have sound understanding of the concept and applications of academic writing
CO2	have acquired enough knowledge of academic writing style, strategy and approach
CO3	be able to demonstrate error free and effective academic writing
CO4	be able to demonstrate ability to work on project/report/paper writing
CO5	understand the concept of plagiarism and learn to use different citation styles as a part of referencing

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-
CO4	3	-	-	-	3	-	-
CO5	3	-	-	-	-	-	2

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

V. Reference Books / Reading**Essential Reading for Concepts**

- Academic Writing for International Students, Routledge
- Academic Writing: A Guide for Management Students and Researchers. Monipally, M. M. & Pawar, B. S. Sage. 2010. New Delhi

Essential Reading for Activity and Teacher Resource

- *Effective Academic Writing Level - 1,2,3,4 (Second Edition)* By: Alice Savage, Patricia Mayer, Masoud Shafiei, Rhonda Liss, & Jason Davis; Publisher: Oxford

Additional Reading

- Writing Your Thesis (2nd Edition) by Paul Oliver, Sage
- Development Communication In Practice by Vilanilam V J, Sage
- Intercultural Communication by Mingsheng Li, Patel Fay, Sage
- www.owl.perdue.edu

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK AND RITA PATEL INSTITUTE OF PHYSIOTHERAPY
 University Elective- Semester II
PT796.01 FITNESS & NUTRITION

CREDIT HOURS:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
-	2	2	-	2	2	-	100	100

A. OUTLINE OF THE COURSE:

Sr. No.	Title of the unit	Minimum number of hours
1.	PHYSICAL EDUCATION AND PHYSICAL FITNESS	10
2.	NUTRITION AND HEALTH	15
3.	SPORTS AND LIFE SKILLS EDUCATION	15

Total hours (Theory): 00

Total Hours (Practical):40

Total Hours: 40

B. DETAILED SYLLABUS:

1	INTRODUCTION TO THE USE OF DRUGS IN SPORT	10 Hours	25%
1.1.	Concept of Physical Education, Meaning, Definition, Aims and Objectives of Physical Education, Need and Importance of Physical Education		
1.2	Physical Education and its Relevance in Inter Disciplinary Context		
1.3	Physical Fitness Components: Type of Fitness, Health Related Physical Fitness, Performance Related Physical Fitness		
2	NUTRITION AND HEALTH	15 Hours	37.5%
2.1	Concept of Food and Nutrition, Balanced Diet, Food Pyramid Index		
2.2	Macro and Micronutrients: Types, functions and classification system		
2.3	Carbohydrates: Types, RDA data, Glycemic index, Sources of Fats, saturated, unsaturated fats, recommended intake, importance of fat in diet, fats in health and disease		
2.4	Protein: Types EAA, function, assessing quality of proteins, selecting incomplete Proteins, RDA sources.		
2.5	Vitamins And Minerals: Types, functions, sources, and minerals - calcium, Phosphorus, iron, magnesium, sodium, potassium, and chloride. Trace elements - Sources and functions.		
2.6	Determining Caloric Intake and Expenditure, Obesity, Causes and Preventing Measures – Role of Diet and Exercise, Importance of hydration in exercise		
3	SPORTS AND LIFE SKILLS EDUCATION	15 Hours	37.5%
3.1	Sports and Socialization		
3.2	Physical Activity and Sport – Emotional Adjustment and Wellbeing		
3.3	Substance Abuse among Youth – Preventive Measures and Remediation		
3.4	Yoga, Meditation and Relaxation		
3.5	Sports and Character Building		
3.6	Values in Sports		
3.7	Sports for World Peace and International Understanding		

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	To underline general concepts of physical education and physical fitness.
CO2	To Appraise the importance of exercise in the maintenance of health and fitness.
CO3	To promote and understand the value of sports for life skill development.

CO4	Introduce nutritional principles and application to improve overall health.
CO5	To apply fitness related components in fitness assessment.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	1	1	1	-	-	-
CO2	1	1	2	3	1	-	-
CO3	1	-	2	3	1	-	-
CO4	1	1	2	3	-	-	-
CO5	1	3	2	1	-	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended study material:

❖ **Textbooks**

1. ACSM’s “Health Related Physical Fitness Assessment Manual Lippincott Williams and Walkins USA, 2005.
2. Siedentop D (1994) Introduction to Physical Education and Sports (2nd ed.), California: Mayfield Publishing Company.
3. Corbin Charles Beetal C.A., (2004) Concepts of Fitness and Welfare Boston McGraw Hill

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester II
PT754 - CLINICAL EDUCATION II

Credit Hours

Teaching Scheme			Examination Scheme					
Contact Hrs.			Credit	Theory		Practical		Total
Theory	Practical	Total		Internal	External	Internal	External	
-	16	16	8	-	-	60	140	200

Total Hours: 320 Hours

A. Outline of the course:

Sr. No.	Skill Sets
1	Display professional and ethical behavior while in the assigned treatment environment.
2	Demonstrate effective and appropriate oral, nonverbal and written communication skills.
3	Assure safety of patient and self throughout patient care.
4	Assure patient privacy, rights, and dignity.
5	Examination • Muscle Strength • Muscle length • Special test to identify various conditions • Coordination • Posture • Balance / fall risk assessment • Gait • Activities of daily living • Pain • Environmental assessment • Differential diagnosis for various conditions
6	Provide competent provision of physiotherapy interventions, including: Therapeutic exercise A. Aerobic Capacity/Endurance Conditioning or Reconditioning B. Balance, coordination, and agility training C. Body mechanics and postural stabilization D. Flexibility exercises E. Gait and locomotion training F. Relaxation G. Strength, power, and endurance training for head, neck, limb, trunk, and ventilatory muscles

7	Functional training in self-care and home management A. Activities of daily living (ADL) training B. Device and equipment use and training C. Injury Prevention or reduction
8	Manual therapy techniques
9	Use clinical problem-solving skills in patient care.
10	Anthropometric characteristics
11	Problem focused management skills (Balance, strength, function, posture, mobility, standing and walking)
12	Assessment and Evaluation Skills to record prognosis or progression
13	Evidence based practice to the development and improvement of clinical practice
14	Case presentation of physical findings, evaluation, and diagnosis, patient education and therapeutic plan
11	Professional Commitment and Attitude Towards improving the quality of care

Clinical training comprises all of the formal and practical "real-life" learning experiences provided for students to apply knowledge and skills in the clinical environment. Experiences would include those of short and long duration (e.g., part-time, full-time) and those that provide a variety of learning experiences (e.g., rotations on different units within the same practice setting such as Musculoskeletal Physiotherapy OPD & IPD etc.) to include comprehensive care of patients. Each student will be under the supervision of a clinical supervisor at the clinical education site who directly instructs and supervises students during their clinical learning experiences. Clinical supervisors will evaluate each student twice during clinical training (mid-term and final) using a standard evaluation tool.

The student will be formally evaluated through **Objective Structured Clinical Examination (OSCE)** by the internal and external examiners.

A. Course Outcomes:

At the end of the course, the students will be able to

CO1	Demonstrate communication skills that facilitate empathetic relationships and effective collaborations with patients, caregivers and other health care professionals.
CO2	Demonstrate skills of history taking from patient in more complex conditions.
CO3	Demonstrate understanding of the elements of a comprehensive Musculoskeletal physiotherapy assessment for various conditions
CO4	Organize case presentation to reflect details of physical findings, reasons for evaluation, and justify the diagnostic, patient education and therapeutic plan.
CO5	Conduct SMART goal setting with appropriate evidenced based interventions for patient in conjunction with their family and community
CO6	Demonstrate commitment to achieving personal and professional excellence in managing persons with various musculoskeletal conditions.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	-	-	1	2
CO2	-	3	2	-	-	-	-
CO3	-	2	3	-	-	-	-
CO4	2	-	-	2	-	3	-
CO5	-	-	-	-	3	-	2
CO6	-	-	-	-	-	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester II
PT797 BASIC SCIENCES (BIOMECHANICS, PATHOMECHANICS & ERGONOMICS)

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
4	-	4	4	-	4	100	-	100

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	SKELETAL CONSIDERATION OF MOVEMENT	09
2.	NEUROMUSCULAR CONSIDERATION OF MOVEMENT	10
3.	KINETICS & KINEMATICS	20
4.	PATHOMECHANICS	25
5.	ERGONOMICS	16

Total hours (Theory): 80

Total hours (Practical): 00

Total hours: 80

B. Detailed Syllabus:

1	SKELETAL CONSIDERATION OF MOVEMENT	09 Hours	11.25%
1.1	Functions of skeletal system.		
1.2	Types of bones.		
1.3	Biomechanical Characteristics of bones: Bone tissue, architecture of bone, strength and stiffness of bone, Types of load, Bony articulations, Types of joints in detail.		
2	NEURO - MUSCULAR CONSIDERATION OF MOVEMENT	10 Hours	12.5%
2.1	Basic nerve structure, Motor unit, Receptors & Synapse, Reflex movement & Volitional movement, Reciprocal innervation & Inhibition.		
2.2	Structure and function of muscle, physical properties of muscle, muscle attachment, functional characteristics of muscles, muscle fiber potential, mechanical components in the muscle.		
2.3	Muscle action and factors influencing muscle force.		
2.4	Action potential, Evoke potential, Kinetic potential, Torque, Power, Strength & Endurance.		
3	KINETICS & KINEMATICS	20 Hours	25%
3.1	Biomechanics of joints of extremities.		
3.2	Biomechanics of Thoracic cage, Biomechanics of Respiration & Circulation.		
3.3	Biomechanics of upper and lower spine with respective girdle.		
3.4	Kinetics / Kinematics of extremity & spinal joints including jogging, running, climbing up and down & A.D.L.		
3.5	Methods of kinetic and kinematic Investigations.		
3.6	Kinetics and kinematic analysis of normal gait.		
4	PATHOMECHANICS	25 Hours	31.25%
4.1	Regional dysfunctions: Muscle dysfunctions, Joint dysfunctions, Nerve dysfunctions.		
4.2	Pathological posture & Pathological gait.		
5	ERGONOMICS	16 Hours	20%

5.1	Work capacity analysis role of physiotherapy industrial setup, job site disability, pre-employment screening, workers functional capacity assessment, work hardening programme industrial therapy.
5.2	Postural examination, job task analysis, educational programme for prevention of injury adult education, documentation, analysis of functional hazards related to Environment / Industry.

Course Outcomes (COs):

CO1	Understand the skeletal and neuro-muscular system considered for movement
CO2	Explain the biomechanical principles related to human movement
CO3	Articulate the concept of pathomechanics in musculoskeletal conditions
CO4	Apply knowledge of ergonomics to effectively communicate, educate the workers

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	-	-	3	-	-	-
CO2	-	-	-	3	-	-	-
CO3	-	3	3	-	-	-	-
CO4	3	3	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Textbooks:

1. Joint structure and function- Cynthia Norkins, 4th Edition, Jaypee Publication.
2. Pathomechanics – Steindler

❖ Reference Books:

1. Biomechanical basis of human movement, Joseph Hamill & Kathleen M.Knutzen, 3rd Edition, LWW Publications.
2. Bio-mechanics of Musculoskeletal System by Nigg, 2nd Edition, John Wiley

❖ Publication.

1. Kinesiology by K Wells, 6th Edition; Saunders Publication.
2. Clinical kinesiology – Brunnstorm, 5th Edition, Jaypee Publication.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester II
PT798 EXERCISE PHYSIOLOGY & NUTRITION

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
5	-	5	5	-	5	100	-	100

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	PRINCIPLES OF EXERCISE PHYSIOLOGY	10
2.	EXERCISE TESTING AND PLANNING	25
3.	EXERCISE PRESCRIPTION AND TRAINING	25
4.	ENERGY TRANSFER/ DELIVERY SYSTEM	10
5.	ENERGY EXPENDITURE	10
6.	MEASUREMENT AND ASSESSMENT	10
7.	COMPONENTS OF NUTRITION	10

Total hours (Theory): 100

Total hours (Practical): 00

Total hour: 100

B. DETAILED SYLLABUS:

1	PRINCIPLES OF EXERCISE PHYSIOLOGY	10 Hours	10%
1.1	Role of Aerobic and Anaerobic mechanism during exercises.		
1.2	Acute effects of High, Burst and Short duration exercises.		
1.3	Acute effect of Steady level exercise on following parameters – Blood flow, Heart rate, Blood Pressure, Pulse Rate, Respiration Rate, Acid Base Balance, Body Temperature, Fluid-Electrolyte Balance and Substrate Utilization.		
1.4	Aging and physiologic function, physical activity in the different types of population.		
1.5	Clinical exercise physiology for cancer, cardiovascular and pulmonary rehabilitation.		
2	EXERCISE TESTING AND PLANNING	25 Hours	25%
2.1	Definition of exercise testing, the need for exercise testing, clinical assessment of exercise tolerance, factors affecting exercise tolerance, diagnostic use of exercise testing, indications of exercise testing, exclusion criteria, Clinical values of exercise testing.		
2.2	Objective assessment of exercise- related symptoms. Muscles fatigue and weakness, dyspnea, exertional chest pain, Applications in cardiac disorders, coronary artery disease, congenital heart disease, valvular disease, pulmonary vascular disease, peripheral vascular disease.		
2.3	Low Level Exercise Testing: Purpose, Contra - indications, Termination points.		
2.4	Maximal Exercise Testing: Purpose, Guidelines, Exercise test protocols, Contraindications and Precautions, Criteria for termination of test, Prognostic implications from exercise testing.		
2.5	Exercise electrocardiogram. The normal ECG. Normal response disease, changes associated with coronary disease. Changes in the ST segment and T waves, Effects of drugs, cardiac arrhythmias in exercise, abnormalities of conduction, the ECG in athletes.		
2.6	Approaches to clinical exercise testing, master step test, the balke protocol, the Bruce protocol, Scandinavian protocol, triangular protocol, walking protocol, Wingate test, maximal oxygen uptake, the stage I test, stage 2,3 and 4 tests, indications and contraindications to exercise testing.		
2.7	Exercise tolerance test or stress test METS and their use in evaluation.		
3	EXERCISE PRESCRIPTION & TRAINING	25 Hours	25%
3.1	Introduction to exercise prescription, the individual approach, the aerobic session, frequency, time, mode of exercise, rate of progression, musculoskeletal conditioning, static stretching, systems of muscular strength and endurance training, Recommendation based on maximal exercise test results, interpretation of maximal exercise test results.		
3.2	Conditioning effects of various levels of Sub-Maximal Exercises. Considerations of age and sex in		

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3.3	Exercise prescription for health and fitness with special emphasis to cardiovascular disease, Obesity and Diabetes, Fatigue – Types, Relevance with Exercise Tolerance tests & Training. Principles of health promotion for Growing Children, Healthy Adults, Pregnant/Lactating females, Elderly, Sports person. Special aids to exercise training and performance.		
3.4			
3.5			
4	ENERGY TRANSFER/DELIVERY SYSTEM	10 Hours	10%
4.1	Introduction to energy transfer, energy transfer in the body-phosphate bond energy, energy released from food, energy transfer in exercise. Systems of energy delivered and utilization: Cardiovascular system: cardiovascular regulation, integration and functional capacity. Pulmonary System: Dynamics of pulmonary ventilation, regulation of pulmonary ventilation, pulmonary ventilation during exercise, acid-base regulation. Endocrine system: organization, acute and chronic response to exercise. Role of Musculoskeletal system & Nervous System		
4.2			
4.2.1			
4.2.2			
4.2.3			
4.2.4			
5	ENERGY EXPENDITURE	10 Hours	10%
5.1	Expenditure during rest, confinement during illness and various levels of Physical Exercises, factors influencing energy uptake and substrate utilization. Measurement of Human energy expenditure, individual differences and measurement of energy capacities. Energy expenditure during walking, jogging, running and swimming.		
5.2			
5.3			
6	MEASUREMENT AND ASSESSMENT	10 Hours	10%
6.1	Body composition assessment, physique, performance, and physical activity, over-weight, obesity and weight control. Lab Investigations - blood glucose, lipid profile, electrolytes, hemoglobin.		
6.2			
7	COMPONENTS OF NUTRITION	10 Hours	10%
7.1	Carbohydrates: Types, RDA data, Glycemic index, Sources of Fats, saturated, Unsaturated fats, recommended intake, importance of fat in diet, fats in health and disease. Protein: Types EAA, function, assessing quality of proteins, selecting incomplete proteins, RDA sources. Vitamins And Minerals: Types, functions, sources, and minerals- calcium, Phosphorus, iron, magnesium, sodium, potassium and chloride. Trace elements- Sources and functions. Nutritional status in India, RDA by ICMR, basic five food groups, Principles of Menu planning, Nutritional deficiencies-protein calorie malnutrition, effects of Nutrition on teeth, role of fluoride.		
7.2			
7.3			
7.4			

7.5	Physical Activity, Aging & Sports, Obesity–prescription of Diet & its modification. Role of nutrient inhibitors. Effect of Food toxins and non-nutritive foods. Food Contamination and spread of diseases.
7.6	Types of diet–veg/non-veg. Advantages and disadvantages of the same, Food Processing, methods of food preparation.
7.7	

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Understand exercise physiology in healthy and diseased population
CO2	Understand body composition measurement, exercise and related symptoms testing in healthy and pathological conditions
CO3	Understand exercise prescription for healthy and various common diseased population
CO4	Understand components and types of diet for various population

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	-	-	3	-	-	-
CO2	-	3	3	-	-	-	-
CO3	-	3	3	3	-	-	-
CO4	-	-	3	3	-	-	-

Recommended Study Material:

❖ Textbooks:

1. Katch: Exercise physiology, energy nutrition and human performance.
2. Scott K Powers: Theory and application to fitness and performance.

❖ Reference Books:

1. Axen: Illustrated principles of exercise physiology.
2. Frank: Exercise physiology for health care professionals.
3. Tudor Hale: Exercise physiology – a thematic approach.
4. George Brooks: Exercise physiology -Human bioenergetics and its application.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester II
PT868 EXERCISE PROGRAMMING

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
3	2	5	3	1	4	50	50	100

A.Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction To Exercise Prescription	03
2.	Cardiovascular Fitness	10
3.	Flexibility And Balance Testing	05
4.	Muscular Strength And Endurance Testing	05
5.	Body Composition	09
6.	Exercise Planning And Prescription For Selected Populations	20
7.	Alternative Training Modalities:	08

Total hours (Theory) : 60

Total hours (Practical): 40

Total hours 100

B.Detailed Syllabus:

1.	Introduction To Exercise Prescription	03 Hours	03%
1.1	Purpose		
1.2	Components		
1.3	Screening and Risk Stratification		
2.	Cardiovascular Fitness	10 Hours	10%
2.1	Type of Activity		
2.2	Classification of Endurance Activities		
2.3	Advantages and Disadvantages Intensity		
2.4	of Conditioning		
	Percent of Maximum Heart Rate		
	Percent of Maximum Oxygen Consumption		
	Rate of Perceived Exertion		
	Karvonen's Method of Using Heart Rate		
	Talk Test		
2.5	METs		
2.6	Duration		
2.7	Frequency		
	Rate of Progression		
	Initial Conditioning Stage Improvement Conditioning StageMaintenance Conditioning Stage		
3.	Flexibility And Balance Testing	05 Hours	05%
3.1	Dynamic Procedures		
3.2	Static Procedures		
3.3	Proprioceptive Neuromuscular Facilitation (PNF) Procedure		
4.	Muscular Strength And Endurance Testing	05 Hours	05%
4.1	Isometric Training		
4.2	Isotonic		
4.3	Training		
	Isokinetic		
	Training		
5.	Body Composition	09 Hours	09%
5.1	Body composition Analysis		
5.2	Nutritional		
5.3	ConsiderationsWeight		
5.4			

	Management Designing weight gain programs		
6	Exercise Planning And Prescription For Selected Populations	20 Hours	20%
6.1	Coronary Heart Disease[Phase I, II, and III (In Patient, Out Patient)]		
6.2	Angina, Pacemakers, Percutaneous Transluminal Coronary Angioplasty, Transplants,		
6.3	Chronic Heart Failure or Severe Left Ventricular Dysfunction, Myocardial Infarction		
6.4	Chronic Obstructive Pulmonary Disease: Disease Pathology & Exercise Considerations		
6.5	Diabetes: Disease Pathology: Type I and II, Purpose of Diabetes, Management, Exercise Prescription Considerations,		
6.6	Complications: Hypoglycemia, Hyperglycemia, Ketoacidosis		
6.7	Peripheral Vascular Disease		
6.8	Arthritis Osteoporosis		
6.9	Pregnancy		
6.10	Geriatric population Renal disease		
7.	Alternative Training Modalities:	08 Hours	08%
7.1	Yoga with Sun Salutations		
7.2	Foam Roller		
7.3	Stability Ball		

Course Outcomes:

At the end of the course, the students will be able to

CO1	Able to comprehend the diverse content of exercise physiology and how it applies to physical fitness testing and programming.
CO2	Perform and interpret specific tests of physical fitness in laboratory and field settings.
CO3	Develop the knowledge and skills to evaluate the risks and associated responsibilities then conducting pre-exercise screening, exercise testing and exercise programming.
CO4	Able to comprehend the importance and potential for physical activity/exercise training in the prevention and management of diseases such as heart disease, cancer, diabetes and obesity etc.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	-	-	-	-	-	-
CO2	-	3	1	-	-	-	-
CO3	-	3	2	1	-	-	-
CO4	-	3	2	1	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:**❖ Textbooks:**

1. American College of Sports Medicine. (1993). Resource Manual for Exercise Testing and Prescription. (8th ed.) Philadelphia: Williams & Wilkins.
2. Advanced Fitness Assessment and Exercise Prescription (1994), 7th ed. Heyward and Gibson.
3. Physical Rehabilitation (5th ed) O’Sullivan S B

❖ Reference Books:

1. Exercise Testing and Prescription (1999). Nieman, D.C.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester II
PT963.01 LITERATURE REVIEW

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
3	2	5	3	1	4	100	100	200

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	INTRODUCTION TO LITERATURE REVIEW	10
2.	ACCESSING LITERATURE	15
3	SUMMARISING AND SYNTHESISING INFORMATION	15
4	WRITING UP THE LITERATURE REVIEW	20

Total hours (Theory) : 60 Hrs.

Total hours (Practical): 40 Hrs.

Total hours: 100 Hrs.

B. Detailed Syllabus:

1	INTRODUCTION TO LITERATURE REVIEW	10 Hours	17%
1.1	Introduction and Definition		
1.2	Annotated bibliography vs. literature review		
1.3	Purpose of a literature review		
1.4	Types : narrative review and systematic review		
2	ACCESSING LITERATURE	15 Hours	25%
2.1	Formulating the search question		
2.2	Developing key words and search strategy		
2.3	PubMed Search: Using MeSH terms, advance search, use of limits and filters		
3	SUMMARISING AND SYNTHESISING INFORMATION	15 Hours	25%
3.1	Critical appraisal of the literature		
3.2	Structuring the literature review: The funnel method		
3.3	Coding and creating a matrix		
3.4	Analyzing the literature: Meta-analysis, Meta ethnography and thematic analysis		
4	WRITING UP THE LITERATURE REVIEW	20 Hours	33%
4.1	Sources of bias in creating and completing a literature review.		
4.2	Mistakes commonly made in reviewing research literature		
4.3	Criteria for good literature review		
4.4	Avoiding plagiarism and misinterpretation		

Course Outcome (COs):

At the end of the course, the student will be able to

CO1	Formulate an appropriate review question
CO2	Develop literature search strategy relevant to the review question
CO3	To critically appraise literature in the context of its contribution to understanding the research problem being studied.
CO4	Demonstrate how to construct and write a literature review

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	1	2	1	-	2	3
CO2	-	1	3	-	1	2	1
CO3	1	1	2	1	3	2	2
CO4	-	-	1	2	2	3	-

Recommended Study Material:

❖ Textbooks:

1. Aveyard H. Doing a Literature Review in Health and Social Care: A practical guide (3rd ed). Open University Press; 2014

❖ Reference Books:

1. Ridley D. The literature review: A step-by-step guide for students. Sage Publications; 2012 Jul 23.
2. Booth A. Systematic approaches to a successful literature review. Sage Publications; 2012.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester II
PT870 – NEONATAL CARE

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
3	2	5	3	1	4	100	100	200

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	INTENSIVE CARE OF NEONATE	11
2.	MEDICO-SURGICAL INVESTIGATIONS & PROCEDURES	15
3	ASSESSMENT	8
4	HIGH DEPENDENCY CARE OF NEONATE	8
5.	INFECTIONS IN NEONATES	10
6.	MANAGEMENT OF MORBIDITIES IN NEONATES	20
7.	FAMILY CENTERED CARE	8

Total hours (Theory): 48 Hrs.

Total hours (Practical): 32 Hrs.

Total hours: 80 Hrs.

B. Detailed Syllabus:

1	INTENSIVE CARE OF NEONATE	11 Hours	15%
1.1	Environment of NICU		
1.2	Early Intervention Services		
1.3	Potential complications of the NICU		
1.4	NICU experience and its relationship to Sensory Integration		
1.5	Transport of Neonates		
2	MEDICO-SURGICAL INVESTIGATIONS & PROCEDURES	15 Hours	18%
2.1	Resuscitation & Intubation		
2.2	Lumbar puncture, suprapubic tap, Chest tube placement, Genetic testing, Cerebral Function Monitor,		
2.3	Sonography, Urine tests		
2.4	Laser therapy, Phototherapy, ECMO		
2.5	Repair of esophageal atresia, gastroschisis, omphalocele, congenital diaphragmatic hernia, colostomy		
3	ASSESSMENT	8 Hours	10%
3.1	Developmental assessment		
3.2	Preterm & Prematurity developmental assessment		
3.3	Nutrition and feeding screening		
4	HIGH DEPENDENCY CARE OF NEONATE	8 Hours	10%
4.1	Problems of prematurity		
4.2	Theoretical perspectives for developmental supportive care		
4.3	Pain Management		
4.4	Palliative Care		
5	INFECTIONS IN NEONATES	10 Hours	12%
5.1	Timing of Infections		
5.2	Types of Neonatal infection		
5.3	Identifying and Treating Minor & Major Infections		
6	MANAGEMENT OF MORBIDITIES IN NEONATES	20 Hours	25%
6.1	Respiratory (RDS, ;BPD, MAS, Apnea of Prematurity, Birth Asphyxia, TTN)		
6.2	Neurological (PVL, HIE, IVH, Encephalopathy, NEC, ROP)		
6.3	Polycythemia, Anemia, Hypotension, Hypoglycemia, IEM, PDA		
7	FAMILY CENTERED CARE	8 Hours	10%
7.1	Professional- Parent Partnerships		

7.2	Parents' responses and Parent-Infant Interaction
7.3	Caregiving and the Environment
7.4	Sleep position protocols

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Demonstrate skills of communication, assessment and management and recognize potential adverse outcomes for the neonate
CO2	Interpret indications, limitations, and gestational-age norms laboratory and imaging studies unique to the NICU setting
CO3	Evaluate, formulate a differential diagnosis with appropriate prioritization and manage common signs and symptoms of disease in high risk newborns
CO4	Demonstrate effective approach to assessment and daily management of seriously ill neonates using decision-making and problem solving skills
CO5	Demonstrate effectively as a part of an interdisciplinary team member in the NICU
CO6	Demonstrate knowledge of ethical and medical-legal consideration in the care of critically ill newborns

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	-	-	-	2	1
CO2	-	3	2	-	-	-	-
CO3	-	-	3	2	-	-	-
CO4	-	-	2	3	-	-	-
CO5	2	-	-	-	3	-	-
CO6	-	-	-	-	-	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Materials:

❖ Textbooks:

1. A Guide to Care and Management of Very Low Birth weight infants – A team approach, Caryl J. Semmler
2. The Premature Baby Book: Helen Harrison
3. Developmental Care of Newborns & Infants : Carole Kenner, Jacqueline M. McGrath
4. Pediatric physical therapy: Jan S. Tecklin

❖ Reference Books:

1. Developmental disabilities in Infancy and Childhood: Arnold J. Capute & Pasquale J. Accardo
2. Neurologic interventions for Physical therapy: Suzanne Martin

3. Neurological Rehabilitation: Umphred, Darcy, A.
4. Physical therapy for Children: Suzann K. Campbell
5. Decision making in Pediatric Neurologic Physical therapy: Suzann K. Campbell

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester III
PT901.01- CLINICAL EDUCATION III

Credit Hours

Teaching Scheme				Examination Scheme				
Contact Hrs.			Credit	Theory		Practical		Total
Theory	Practical	Total		Internal	External	Internal	External	
-	10	10	5	-	-	30	70	100

Total Hours: 200 Hours

A. Outline of the course:

Sr. No.	Skill Sets
1	Display professional and ethical behavior while in the assigned treatment environment.
2	Demonstrate effective and appropriate oral, nonverbal and written communication skills.
3	Assure safety of patient and self throughout patient care.
4	Assure patient privacy, rights, and dignity.
5	Examination • Muscle Strength • Muscle length • Special test to identify various conditions • Coordination • Posture • Balance / fall risk assessment • Gait • Activities of daily living • Pain • Environmental assessment • Differential diagnosis for various conditions
6	Provide competent provision of physiotherapy interventions, including:Therapeutic exercise A. Aerobic Capacity/Endurance Conditioning or Reconditioning B. Balance, coordination, and agility training C. Body mechanics and postural stabilization D. Flexibility exercises

	E. Gait and locomotion training F. Relaxation G. Strength, power, and endurance training for head, neck, limb, trunk, and ventilatory muscles
7	Functional training in self-care and home management A. Activities of daily living (ADL) training B. Device and equipment use and training C. Injury Prevention or reduction
8	Manual therapy techniques
9	Use clinical problem-solving skills in patient care.
10	Common investigative procedures in Musculoskeletal conditions
11	Problem focused management skills (Balance, strength, function, posture, mobility, standing and walking)
12	Assessment and Evaluation Skills to record prognosis or progression
13	Evidence based practice to the development and improvement of clinical practice
14	Education for exercise adherence and home compliance
15	Case presentation of physical findings, evaluation, and diagnosis, patient education and therapeutic plan
16	Working effectively as a member of the rehabilitation team
17	Evaluation for the need of referral services after detail assessment of red flags
18	Professional Commitment and Attitude Towards improving the quality of care

Clinical training comprises all of the formal and practical "real-life" learning experiences provided for students to apply knowledge and skills in the clinical environment. Experiences would include those of short and long duration (e.g., part-time, full-time) and those that provide a variety of learning experiences (e.g., rotations on different units within the same practice setting such as Musculoskeletal Physiotherapy OPD & IPD etc.) to include comprehensive care of patients. Each student will be under the supervision of a clinical supervisor at the clinical education site who directly instructs and supervises students during their clinical learning experiences. Clinical supervisors will evaluate each student twice during clinical training (mid- term and final) using a standard evaluation tool.

The student will be formally evaluated through **Objective Structured Clinical Examination (OSCE)** by the internal and external examiners.

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Demonstrate professional behaviour and respectful communication with patients, caregivers and colleagues in all educational activities.
CO2	Safely administer and interpret standardized assessment procedures with published guidelines for reference.
CO3	Monitor patient response to therapy/exercise and adjust treatment accordingly.
CO4	Identify how to progress the patient towards achieving set goals.
CO5	Identify PT treatments specific to adults with musculoskeletal condition population according to the best evidence-to date
CO6	Identify and recommend appropriate equipment, devices, and orthotics

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	-	-	-	-	-
CO2	-	3	2	-	-	1	-
CO3	-	2	3	-	-	-	-
CO4	-	-	2	2	-	-	-
CO5	-	1	-	3	-	-	2
CO6	-	-	-	-	-	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF
PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester III
PT902.01-DISSERTATION I

INTRODUCTION

The dissertation is aimed to train a postgraduate student in research methods and techniques. It includes identification of a problem, formulation of a hypothesis, search and review of literature getting acquainted with recent advances, designing of a research study, collection of data, critical analysis, and comparison of results and drawing conclusions. Every candidate pursuing MPT degree course is required to carry out work on a selected research project under the guidance of a recognized postgraduate teacher. The result of such a work shall be submitted in the form of dissertation.

Course Work:

Submission of dissertation proposal to the dissertation proposal approval committee

Content of a proposal:

The proposal shall include the following sections.

- 1. Introduction:** This should give a short overview of the problem, issues, or topic to be studied and reasons why it is an important problem to study. Often, it includes some historical background of the problem.
- 2. Statement of the problem:** This is a more complete discussion of the problem of the dissertation. The major questions, hypotheses, or statements should be included with a possible list of secondary questions if appropriate. This section is an extended discussion of the problem often with some literature noted.
- 3. Statement of the significance of this study to the field :** In other words, why is this study important? It should include a statement about how it advances knowledge in the field with reference to past literature and general concerns of the area.
- 4. Literature Review related to the problem, issue, or topic:** In this, various schools of thought on this problem are explored with significant attention to the conceptual and the theoretical aspects of the problem and how those contribute to the topic's development.
- 5. Methodology:** It includes a discussion of the research design, the population to be studied, discussion of appropriate instruments, sampling concerns, data collection approaches, analyses to be conducted and a projected timetable for completion of the dissertation.
- 6. Limitations:** It is to clarify or limit the scope of the method(s) employed in this study.

Once the student has received the permission from their Dissertation guide to defend their proposal, the proposal defense will be scheduled. The proposal approval committee would be

constituted consisting of 3 members with one external member, one internal faculty (other than guide) from the institute and the head of institute. The dissertation committee member's backgrounds should complement the area of anticipated study. This does not require that each member be an expert on every aspect of the study. Potentially, the members will all bring different strengths or knowledge base to the committee.

The dissertation proposal approval committee approves the dissertation proposal subject to the further approval from IRB.

The Institutional Review Board Approval Process:

Researchers who use human participants in their research must follow specific guidelines as a condition for using the data provided by these persons. If the proposal involves human participants, the student will need to submit an application to the Institutional Review Board (IRB) of the Institute. In consultation with his or her Guide, the student prepares an IRB application and submits it to the institute's IRB coordinator. The IRB judge's application on issues related to protection of participants from physical and emotional distress and not on theoretical or methodological grounds. The study cannot go forward until IRB approval is received. In many cases, minor revisions to the study will be necessary to gain approval. Collecting data prior to receiving IRB approval is considered sufficient grounds for halting a research study. In general, students should not submit their IRB application before the proposal defense as any major changes in the proposal would necessitate re-applying to the IRB.

This proposal is considered an agreement or contract describing how the student will conduct the study and cannot be changed without guide/committee approval. Substantial changes to the methods, goals, and objectives articulated in a proposal will necessitate a new proposal defense.

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Recognise the importance of planning and preparation required to undertake a research project
CO2	Articulate a clear research question or problem and formulate a hypothesis
CO3	Know existing body of research relevant to their topic and explain how their project fit
CO4	Critically analyse and evaluate the knowledge and understanding in relation to the agreed area of study
CO5	Design an appropriate research proposal to carry out research

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	2	2	2	-	-	-
CO2	2	2	2	2	-	-	-
CO3	-	-	-	2	2	-	-
CO4	-	-	-	2	2	-	-
CO5	-	-	-	-	-	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester III
PT903.01 PHYSICAL & FUNCTIONAL DIAGNOSIS FOR MUSCULOSKELETAL SCIENCES

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
4	2	6	4	1	5	100	100	200

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum Number of Hours
1.	PRINCIPLES AND CONCEPTS OF ORTHOPEDIC PHYSICAL EXAMINATION	3
2.	TEMPOROMANDIBULAR JOINT AND SPINE	7
3.	UPPER LIMB	7
4.	PELVIS AND LOWER LIMB	7
5.	FRACTURE AND DISLOCATION	5
6.	PAIN	4
7.	ASSESSMENT OF GAIT	4
8.	ASSESSMENT OF POSTURE	3
9.	ASSESSMENT OF THE AMPUTEE	3
10.	PERIPHERAL NERVE INJURIES	5
11.	DISEASES OF BONES & JOINTS	6
12.	INFLAMMATORY AND DEGENERATIVE CONDITIONS	6
13.	DEFORMITIES	6
14.	GERIATRICS	4
15.	SPORTS INJURIES	6
16.	ICF & DISABILITY EVALUATION	4

Total hours (Theory): 80 Hrs.

Total hours (Practical): 40 Hrs.

Total hours: 120 Hrs.

B. Detailed Syllabus:

1	PRINCIPLES & CONCEPTS OF ORTHOPEDIC PHYSICAL EXAMINATION	3 Hours	2.5%
1.1	Patient history.		
1.2	Observation.		
1.3	Palpation		
1.4	Examination: Principles, Scanning examination, Examination of specific joints, Functional assessment, Reflexes & cutaneous distribution, Special tests, Investigation.		
2	TEMPOROMANDIBULAR JOINT AND SPINE	7 Hours	5.8%
2.1	Examination of tempomandibular joint, cervical spine, thoracic spine and lumbar spine.		
2.2	Temporomandibular joint: Temporomandibular joint dysfunction (TMD)		
2.3	Spine: Spondylosis, Spinal canal stenosis, Prolapse intervertebral disc (PVD), Cervical and lumbar radiculopathy, Cervical myelopathy, Facet joint pathology, Vertebral basilar artery problems, Torticollis, Cervical rib, Thoracic outlet syndrome, Brachial neuralgia, Low back pain, Spondylolisthesis, Instability of spine.		
3	UPPER LIMB	7 Hours	5.8%
3.1	Examination of shoulder, elbow, forearm, wrist and hand.		
3.2	Shoulder Complex: Rotator cuff injuries, Frozen shoulder, Instability, Impingement syndrome, Subacromial bursitis, Labral lesions, and Overuse tendinitis.		
3.3	Elbow Complex: Lateral epicondylitis, Medial epicondylitis, and Olecranon bursitis.		
3.4	Forearm, wrist & hand complex: Cubital tunnel syndrome, Pronator teres syndrome, Carpal tunnel syndrome, Guyon canal syndrome, Dupuytren's contracture, Trigger finger, De quervain's tenosynovitis, Flexor tendon injuries, Extensor tendon injuries, Grip strength testing, Pinch strength testing.		
4	PELVIS AND LOWER LIMB	7 Hours	5.8%
4.1	Examination of pelvis, hip, knee, lower leg, ankle and foot.		
4.2	Pelvis and hip joint: Sacroiliac dysfunction, Legg calve perthes disease, Slipped capital femoral epiphysis.		
4.3	Knee joint: Anterior & posterior cruciate ligament injuries, Medial & lateral collateral ligament injury, Meniscal injuries, Patello-femoral disorders, Plica syndrome, Patellar tendon ruptures, Osgood-Schlatter disease.		
4.4	Lower leg, ankle joint & foot: Ankle sprains, Inferior heel pain (plantar fasciitis), Achilles tendon dysfunction, Tarsal tunnel syndrome, Metatarsalgia, Morton's neuroma.		
5	FRACTURE AND DISLOCATION	5 Hours	4.2%
5.1	Fracture: Definition, Types, Signs and symptoms.		
5.2	Fracture healing, Complications of fractures.		

5.3	Principles of management: Conservative & surgical approaches, Reduction (open / close), Immobilization (fixations) etc.		
5.4	Subluxation / dislocations: Definition, Signs & symptoms, Classification, Management (conservative and operative).		
5.5	Physiotherapy assessment of fractures & dislocations of upper limb, spine & lower limb.		
6	PAIN	4 Hours	3.3%
6.1	Historical perspective, Theories of pain, Classification of pain, Clinical manifestations.		
6.2	Patient interview, Body charts and Pathological assessment (SIP, MPI, McGill Pain Questionnaire, VAS, SCL-90 and other methods of pain assessment).		
6.3	Assessment of pain syndromes		
7	ASSESSMENT OF GAIT	4 Hours	3.3%
7.1	Gait cycle, Stance phase, Swing phase, Double-leg stance, Single-leg stance.		
7.2	Normal parameters of gait: Base width, Step length, Stride length, Lateral pelvic shift, Vertical pelvic shift, Pelvic rotation, Center of gravity, Normal cadence.		
7.3	Normal pattern of gait, Joint motion during normal gait.		
7.4	Gait analysis, Locomotion Score, Compensatory Mechanisms.		
7.5	Abnormal Gait: Antalgic (painful) gait, Arthrogenic (stiff hip or knee) gait, Ataxic gait, Contracture gaits, Equinus gait, Gluteus Maximus gait, Gluteus medius (Trendelenburg's) gait, Hemiplegic or hemiplegic gait, Parkinsonism gait, Plantar flexor gait, Psoatic limp, Quadriceps gait, Scissors gait, Short leg gait, Step page or drop foot gait.		
8	ASSESSMENT OF POSTURE	3 Hours	2.5%
8.1	Postural development, Factors affecting posture, Causes of poor posture.		
8.2	Common spinal deformities: Lordosis, Kyphosis, Scoliosis.		
8.3	Methods of Postural analysis.		
9	ASSESSMENT OF THE AMPUTEE	3 Hours	2.5%
9.1	Levels of upper & lower limb amputation, Indications, Complications of amputation.		
9.2	Assessment: Patient history, Observation, Palpation, Examination, Measurements related to amputation, Resisted isometric movements, Functional assessment, Sensation testing, Psychological testing & Investigations.		
10	PERIPHERAL NERVE INJURIES	5 Hours	4.2%
10.1	Brachial plexus injuries including Erb's palsy and Klumpke's palsy.		
10.2	Median, Ulnar and Radial nerve injuries.		
10.3	Sciatic, Femoral, Common peroneal and Posterior tibial nerve injuries.		
10.4	Assessment of peripheral nerve injuries.		
11	DISEASES OF BONES & JOINTS	6 Hours	5%

11.1	Infective conditions: Osteomyelitis (Acute / chronic), Brodie’s abscess, TB spine and major joints like shoulder, hip, knee, ankle, elbow.		
11.2	Arthritic conditions: Pyogenic arthritis, Septic arthritis, Syphilytic infection of joints.		
11.3	Bone Tumors: Classification, clinical features, management - medical and surgical of the following tumors: Osteoma, Osteosarcoma, Osteochondroma, Enchondroma, Ewing’s sarcoma, Giant cell tumor, Multiple myeloma, Metastatic tumors.		
11.4	Perthes disease, Slipped Capital Femoral Epiphysis and Avascular Necrosis.		
11.5	Metabolic Bone Diseases: Rickets, Osteomalacia, Osteopenia, Osteoporosis.		
11.6	Physiotherapy assessment of the diseases of bones and joints		
12	INFLAMMATORY AND DEGENERATIVE CONDITIONS	6 Hours	5%
12.1	Osteoarthritis, Rheumatoid arthritis, Ankylosing spondylitis, Gouty arthritis, Psoriatic arthritis, Hemophilic arthritis, Still’s disease (JRA), Charcot’s joints.		
12.2	Connective Tissue Disorders: Systemic Lupus Erythematosis, Scleroderma, Dermatomyositis.		
12.3	Physiotherapy assessment of the inflammatory and degenerative conditions.		
13	DEFORMITIES	6 Hours	5%
13.1	Congenital Deformities: CTEV, Congenital flat foot (Vertical talus), CDH, Sprangel’s shoulder, Madelung deformity, Radioulnar synostosis, Syndactyly, Polydactyly, Ectrodactyly, Torticollis, Klippel Feil syndrome, Cervical rib.		
13.2	Acquired Deformities: Acquired Torticollis, Scoliosis, Kyphosis, Lordosis, Cova vara, Genu varum, Genu valgum, Genu recurvatum, Pes Cavus, Hallux rigidus, Hallux valgus, Hammer toe, Metatarsalgia.		
13.3	Physiotherapy assessment of the various deformities.		
14	GERIATRICS	4 Hours	3.3%
14.1	Definition of Elderly, Classification of Elderly.		
14.2	Physiological response to ageing: Normal aging, Theories of aging, Features of normal aging (Musculoskeletal system, cardiovascular system, pulmonary function, nervous system, special senses, vestibular system, changes in other systems).		
14.3	Principles of geriatric assessment: Components of geriatric assessment (History, Physical examination, Functional status, Mental status, Emotional status, Laboratory testing).		
14.4	Musculoskeletal disorders in elderly: Osteoporosis, Osteoarthritis, RA		
14.5	Falls in elderly.		
15	SPORTS INJURIES	6 Hours	5%
15.1	Classification of sports injuries, Biomechanics of common sports injuries.		
15.2	Principles of diagnosis of sports injuries: Clinical assessment, Biomechanical examinations.		

15.3	Musculoskeletal fitness: Muscular fitness, Tests for muscular endurance and strength, Flexibility testing.		
16	ICF & DISABILITY EVALUATION	4 Hours	3.3%
16.1	International Classification of Functioning, Disability and Health (ICF).		
16.2	Clinical examination in general and detection of movement dysfunction.		
16.3	Physical disability evaluation & Disability diagnosis.		
16.4	Causes of various locomotor disabilities in different age groups with identification & assessment.		

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Demonstrate skills related to communication, assessment and management and recognize potential adverse outcomes for the Musculoskeletal disorders.
CO2	Interpret the clinical findings, laboratory report and imaging studies based on the evidence
CO3	Integrate diagnostic skill in clinical assessment and management of Musculoskeletal Disorders and diseases
CO4	Demonstrate as an effective interdisciplinary team member and communicate effectively with various stake holders involved in Musculoskeletal Care
CO5	Demonstrate knowledge of ethical and medico-legal consideration, professional responsibility in the assessment and diagnosis of Musculoskeletal Disorders

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	-	-	-	-	-
CO2	-	2	3	-	-	-	-
CO3	-	-	3	3	-	-	-
CO4	-	-	-	-	3	-	-
CO5	-	-	-	-	-	3	3
CO6	-	-	-	-	-	-	-

Recommended Study Material:

❖ Textbooks:

1. Orthopedic Physical Assessment by David J. Magee, 6th Edition.
2. Adams's Outline of Fractures, Including Joint Injuries, 12th Edition.
3. Physical Rehabilitation by Susan B. O'Sullivan and Thomas J. Schmitz, 6th Edition
4. Clinical Sports Medicine by Brukner and Khan, 4th Edition.

5. Principles of Geriatric Physiotherapy by Multani.

❖ **Reference Books:**

1. Pathology and Intervention in Musculoskeletal Rehabilitation by David J. Magee, James E. Zachazewski, and William S. Quillen, 2nd Edition.
2. Textbook of Orthopedics with Clinical Examination Methods in Orthopedics by Ebnezar, 4th Edition.
3. Apley's System of Orthopaedics and Fractures by Louis Soloman, David Warwick, Selvadurai Nayagam, 9th Edition.
4. Exercise Testing and Prescription: A Health-Related Approach by David Nieman, Seventh Edition.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester III
PT904.01 ADVANCED PHYSIOTHERAPEUTICS FOR MUSCULOSKELETAL SCIENCES

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
3	2	5	3	1	4	100	100	200

A. Outline of the course:

Sr. No.	Title of the unit	Minimum Number of Hours
1.	TEMPOROMANDIBULAR JOINT AND SPINE	03
2.	UPPER LIMB	05
3.	PELVIS AND LOWER LIMB	05
4.	FRACTURE AND DISLOCATION	04
5.	PAIN	03
6.	GAIT	04
7.	POSTURE	03
8.	AMPUTATIONS	02
9.	PERIPHERAL NERVE INJURIES	04
10.	DISEASES OF BONES & JOINTS	04
11.	INFLAMMATORY AND DEGENERATIVE CONDITIONS	04
12.	DEFORMITIES	04
13.	GERIATRICS	04
14.	SPORTS INJURIES	05
15.	MISCELLANEOUS	06

Total hours (Theory): 60 Hrs.

Total hours (Practical): 40 Hrs.

Total hours: 100 Hrs.

B. Detailed Syllabus:

1	TEMPOROMANDIBULAR JOINT AND SPINE	3 Hours	5%
1.1	Temporomandibular joint: Conservative, Surgical management and Advanced Physiotherapy treatment of Temporomandibular joint dysfunction (TMD)		
1.2	Spine: Conservative, Surgical management & Advanced Physiotherapy treatment of: Spondylosis, Spinal canal stenosis, Prolapse intervertebral disc (PIVD), Cervical and lumbar radiculopathy, Cervical myelopathy, Facet joint pathology, Vertebral basilar artery problems, Torticollis, Cervical rib, Thoracic outlet syndrome, Brachial neuralgia, Low back pain, Spondylolisthesis and Instability of spine.		
2	UPPER LIMB	5 Hours	8%
2.1	Shoulder Complex: Conservative, Surgical management & Advanced Physiotherapy treatment of: Rotator cuff injuries, Frozen shoulder, Instability, Impingement syndrome, Subacromial bursitis, Labral lesions, and Overuse tendinitis.		
2.2	Elbow Complex: Conservative, Surgical management & Advanced Physiotherapy treatment of Lateral epicondylitis, Medial epicondylitis, and Olecranon bursitis.		
2.3	Forearm, wrist & hand complex: Conservative, Surgical management & Advanced Physiotherapy treatment of: Cubital tunnel syndrome, Pronator teres syndrome, Carpal tunnel syndrome, Guyon canal syndrome, Dupuytren's contracture, Trigger finger, De quervain's tenosynovitis and Tendon injuries.		
3	PELVIS AND LOWER LIMB	5 Hours	8%
3.1	Pelvis and hip joint: Conservative, Surgical management & Advanced Physiotherapy treatment of: Sacroiliac dysfunction, Legg Calve Perthes disease, Slipped capital femoral epiphysis.		
3.2	Knee joint: Conservative, Surgical management & Advanced Physiotherapy treatment of: Anterior & posterior cruciate ligament injuries, Medial & lateral collateral ligament injury, Meniscal injuries, Patello-femoral disorders, Plica syndrome, Patellar tendon ruptures, Osgood-Schlatter disease.		
3.3	Lower leg, ankle joint & foot: Conservative, Surgical management & Advanced Physiotherapy treatment of: Ankle sprains, Inferior heel pain (plantar fasciitis), Achilles tendon dysfunction, Tarsal tunnel syndrome, Metatarsalgia, Morton's neuroma.		
4	FRACTURE AND DISLOCATION	4 Hours	7%
4.1	Fracture: Advanced Physiotherapy management of fracture: Upper limb, lower limb and spine.		
4.2	Subluxation / dislocations: Advanced Physiotherapy management of subluxation/ dislocation of: Upper limb and lower limb		
5	PAIN	3 Hours	5%

5.1	Pain Management for acute and chronic pain syndrome		
6	GAIT	4 Hours	6%
6.1	Rehabilitation of Abnormal Gait in Musculoskeletal Conditions due to: Pain, Muscular weakness/paralysis, Leg length discrepancy.		
7	POSTURE	3 =Hours	5%
7.1	Postural re-education		
7.2	Advanced Physiotherapy management of Common spinal Deformities: Lordosis, Kyphosis, and Scoliosis.		
8	AMPUTATIONS	3 Hours	5%
8.1	Advanced Physiotherapy treatment of various levels of amputation.		
8.2	Pre-amputation, post-amputation (pre-prostheses & prostheses) rehabilitation of upper limb and lower limb amputation.		
9	PERIPHERAL NERVE INJURIES	4 Hours	7%
9.1	Brachial plexus injuries including Erb's palsy and Klumpke's palsy. Median,		
9.2	Ulnar and Radial nerve injuries.		
9.3	Sciatic, Femoral, Common peroneal and Posterior tibial nerve injuries.		
9.4	Conservative, Surgical management and Advanced physiotherapy management of peripheral nerve injuries.		
10	DISEASES OF BONES & JOINTS	4Hours	6%
10.1	Infective conditions: Osteomyelitis (Acute / chronic), Brodie's abscess, TB spine and major joints like shoulder, hip, knee, ankle, elbow.		
10.2	Arthritic conditions: Pyogenic arthritis, Septic arthritis, Syphilitic infection of joints.		
10.3	Bone Tumors: Classification, clinical features, management - medical and surgical of the following tumors: Osteoma, Osteosarcoma, Osteochondroma, Enchondroma, Ewing's sarcoma, Giant cell tumor, Multiple myeloma, Metastatic tumors.		
10.4	Perthes disease, Slipped Capital Femoral Epiphysis and Avascular Necrosis. Metabolic		
10.5	Bone Diseases: Rickets, Osteomalacia, Osteopenia, Osteoporosis. Conservative, Surgical		
10.6	management and Advanced physiotherapy management of the diseases of bones and joints.		

11	INFLAMMATORY AND DEGENERATIVE CONDITIONS	4 Hours	6%
11.1	Osteoarthritis, Rheumatoid arthritis, Ankylosing spondylitis, Gouty arthritis, Psoriatic arthritis, Hemophilic arthritis, Still's disease (Juvenile rheumatoid arthritis), Charcot's joints.		
11.2	Connective Tissue Disorders: Systemic Lupus Erythematosus, Scleroderma, Dermatomyositis.		
11.3	Conservative, Surgical management and Advanced physiotherapy management of the inflammatory and degenerative conditions.		
12	DEFORMITIES	4 Hours	7%

12.1	Congenital Deformities: CTEV, Congenital flat foot (Vertical talus), CDH, Sprangel's shoulder, Madelung deformity, Radioulnar synostosis, Syndactyly, Polydactyly, Ectrodactyly, Torticollis, Klippel Feil syndrome, Cervical rib.		
12.2	Acquired Deformities: Acquired Torticollis, Scoliosis, Kyphosis, Lordosis, Coxa vara, Genu varum, Genu valgum, Genu recurvatum, Pes Cavus, Hallux rigidus, Hallux valgus, Hammer toe, Metatarsalgia.		
12.3	Conservative, Surgical management and Advanced physiotherapy management of the various deformities.		
13	GERIATRICS	4 Hours	7%
13.1	Principles of Geriatric Physiotherapy: Geriatric physical therapy in different settings, Geriatric physical therapy program, Goal setting, Therapeutic intervention.		
13.2	Physiotherapy management of musculoskeletal disorders in elderly: Osteoporosis, Osteoarthritis, Rheumatoid arthritis.		
13.3	Falls in elderly: Prevention and treatment.		
14	SPORTS INJURIES	5 Hour	8%
14.1	Principles of injuries prevention: Systemic injury prevention, Taping and bracing, Protective equipment, Appropriate surfaces, Appropriate training, Principles of training, Training methods, Adequate recovery, Psychology and injury prevention, Nutrition and injury prevention.		
14.2	Treatments used for musculoskeletal examination: Minimizing extent of injury, Immobilization and early mobilization, Heat and cold, Electrotherapeutic modalities, Manual therapy.		
14.3	Principles of Rehabilitation: The rehabilitation program, Soft tissue response to injury, Muscle conditioning, Flexibility, Neuromuscular control (proprioception and balance), Functional exercises, Sport skills, Correction of biomechanical abnormalities, Cardiovascular fitness, Progression of rehabilitation, Stages of rehabilitation, Monitoring the rehabilitation program, Psychology and rehabilitation of injury.		
15	MISCELLANEOUS	6 Hours	10%
	Manual therapy intervention for orthopaedic problem:		
15.1	Maitland: Principles and application in spinal and peripheral joint dysfunction		
15.2	Cyriax: History, physical examination-selective tissue tension test, management strategies in spinal and peripheral joint and soft tissue techniques - deep transverse friction massage, manipulation.		
15.3	McKenzie: Classification of spinal pain as adopted by McKenzie-postural, dysfunction and derangement, Assessment and treatment procedures.		
15.4	Neurodynamics and neural tissue mobilization: biomechanics of neural tissue, Principles of subjective, objective, treatment and reassessment in spinal and extremity adverse neural tension disorders.		

15.5	Myofascial release technique: Principles of assessment and treatment, Management of fibromyalgia and trigger point.
15.6	Muscle energy technique: Variations of MET, Treatment using MET.
15.7	Core stabilization training: Fundamental techniques and progressions.

Course Outcomes (COs):

At the end of the course, the student will be able to

CO1	Demonstrate professional behaviour and respectful communication and conduct patient centric physiotherapy assessments of various orthopaedics conditions using best supportive evidence.
CO2	Analyse the relevant clinical finding with regards to contextual, environmental, and personal factors.
CO3	Develop realistic and acceptable therapeutic goals that are patient centric consistent with the needs and goals of the patient.
CO4	To develop protocols for different physiotherapy musculoskeletal conditions and analyse the physiotherapy related problems that are needed for intervention.
CO5	Demonstrate the evidence based advanced physiotherapy management for musculoskeletal conditions.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2	-	-	-	-
CO2	-	3	-	2	-	-	-
CO3	-	-	-	2	1	-	-
CO4	-	-	-	-	-	2	2
CO5	-	-	3	-	-	-	2

Recommended Study Material:

❖ Textbooks:

1. Clinical Orthopaedic Rehabilitation by S. Brent Brozman & Kevin E. Wilk, 2nd edition.
2. Therapeutic Exercise: Foundations and Techniques by Carolyn Kisner and Lynn Allen Colby, 6th edition.
3. Management of Common Musculoskeletal Disorders Physical Therapy Principles and Methods by Darlene Hertling & Randolph M. Kessler, 4th edition.
4. Orthopaedic Physical Therapy by Robert A. Donatelli & Michael J. Wooden, 4th edition.
5. Clinical Sports Medicine by Brukner and Khan, 4th Edition.
6. Physical Rehabilitation by Susan B. O'Sullivan and Thomas J. Schmitz, 6th Edition.
7. Principles of Geriatric Physiotherapy by Multani.
8. Peripheral Manipulation, Maitland, 4th Edition.
9. Vertebral Manipulation, Maitland, 7th Edition.

10. Manual Therapy: NAGs, SNAGs, MWMs, Etc., Brian R Mulligan, 6th Edition.
11. Clinical Neurodynamics: A New System of Neuromusculoskeletal Treatment.
12. Muscle Energy Techniques (Advanced Soft Tissue Techniques). Chaitow.
13. Text book of Orthopaedic Medicine, diagnosis of soft Tissue Lesions, Cyriax James. 8th Edition.

❖ **Reference Books:**

1. Pathology and Intervention in Musculoskeletal Rehabilitation by David J. Magee, James E. Zachazewski, and William S. Quillen, 2nd Edition.
2. Postural Disorders & Musculoskeletal Dysfunction Diagnosis, Prevention and Treatment by Gill Solberg.
3. The Shoulder by Charles A. Rockwood, 4th edition.
4. Textbook of Orthopedics with Clinical Examination Methods in Orthopedics by Ebnezar, 4th Edition.
5. Dutton's Orthopaedic Examination Evaluation and Intervention, by Mark Dutton, 3rd edition.
6. Apley's System of Orthopaedics and Fractures by Louis Soloman, David Warwick, Selvadurai Nayagam, 9th Edition.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Musculoskeletal Sciences) - Semester IV
PT905.01- CLINICAL EDUCATION IV

Credit Hours

Teaching Scheme			Examination Scheme					
Contact Hrs.			Credit	Theory		Practical		Total
Theory	Practical	Total		Internal	External	Internal	External	
-	10	10	5	-	-	30	70	100

Total Hours: 200 Hours

A. Outline of the course:

Sr. No.	Skill Sets
1	Display professional and ethical behavior while in the assigned treatment environment.
2	Demonstrate effective and appropriate oral, nonverbal and written communication skills.
3	Assure safety of patient and self throughout patient care.
4	Assure patient privacy, rights, and dignity.
5	Examination • Muscle Strength • Muscle length • Special test to identify various conditions • Coordination • Posture • Balance / fall risk assessment • Gait • Activities of daily living • Pain • Environmental assessment • Differential diagnosis for various conditions
6	Provide competent provision of physiotherapy interventions, including:Therapeutic exercise A. Aerobic Capacity/Endurance Conditioning or Reconditioning B. Balance, coordination, and agility training C. Body mechanics and postural stabilization D. Flexibility exercises E. Gait and locomotion training

	F. Relaxation G. Strength, power, and endurance training for head, neck, limb, trunk, and ventilatory muscles
7	Functional training in self-care and home management A. Activities of daily living (ADL) training B. Device and equipment use and training C. Injury Prevention or reduction
8	Manual therapy techniques
9	Use clinical problem-solving skills in patient care A. Determine if patient is safe and comfortable with the intervention, and, if not, determine appropriate modifications B. Compare results of intervention to previously collected data and determine if there is progress toward the expectations established by the PT or if the expectations have been met C. Determine if modifications to the interventions are needed to improve patient response
10	Common investigative procedures in Musculoskeletal conditions
11	Problem focused management skills (Balance, strength, function, posture, mobility, standing and walking)
12	Assessment and Evaluation Skills to record prognosis or progression
13	Determine patient's response to the intervention: A. Interview patient and accurately interpret verbal and nonverbal responses B. Identify secondary effects or complications caused by the intervention C. Determine outcome of intervention (positive or negative), including data collection and functional measures
14	Modify interventions to improve patient response. A. Determine modifications that can be made to the intervention within the plan of care B. Communicate with physical therapist when modifications are outside scope of work or personal scope of work C. Select and implement modification D. Determine patient outcomes from the modification
15	Evidence based practice to the development and improvement of clinical practice
16	Education for exercise adherence and home compliance
17	Case presentation of physical findings, evaluation, and diagnosis, patient education and therapeutic plan
18	Working effectively as a member of the rehabilitation team
19	Evaluation for the need of referral services after detail assessment of red flags
20	Professional Commitment and Attitude Towards improving the quality of care

Clinical training comprises all of the formal and practical "real-life" learning experiences provided for students to apply knowledge and skills in the clinical environment. Experiences would include those of short and long duration (e.g., part-time,

full-time) and those that provide a variety of learning experiences (e.g., rotations on different units within the same practice setting such as Musculoskeletal Physiotherapy OPD & IPD etc.) to include comprehensive care of patients. Each student will be under the supervision of a clinical supervisor at the clinical education site who directly instructs and supervises students during their clinical learning experiences. Clinical supervisors will evaluate each student twice during clinical training (mid- term and final) using a standard evaluation tool.

The student will be formally evaluated through **Objective Structured Clinical Examination (OSCE)** by the internal and external examiners.

Course Outcomes (CSO):

At the end of the course, the students will be able to

CO1	Analyse relevant muscle, joint impairment and associated physical examination to identify impairments contributing to activity limitation and functional limitation.
CO2	Select and justify relevant outcome measures to benchmark recovery.
CO3	Interpret and categorize assessment findings to formulate a reasonable musculoskeletal physiotherapy diagnosis, recognising the factors to consider when predicting recovery for people with musculoskeletal conditions
CO4	Critique selected current musculoskeletal rehabilitation strategies using effective problem solving and research strategies.
CO5	Apply concepts of health care maintenance, conditioning and promotion of health for people with a musculoskeletal problem across the continuum of care.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	1	3	-	-	-	-	-
CO2	-	1	2	-	-	1	-
CO3	-	3	2	-	1	-	-
CO4	-	-	3	2	-	-	-
CO5	-	-	-	3	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

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PT906.01- DISSERTATION II

Course work:

Submission of the completed dissertation

Content :

Every candidate shall submit to the Registrar (Academic) of the university in the prescribed proforma, within 1 month from the date of commencement of the course on or before the dates notified by the university.

The dissertation should be written under the following headings.

1. Introduction
2. Aims or objectives of study
3. Review of literature
4. Methodology
5. Results
6. Discussion
7. Conclusion
8. Summary
9. References
10. Tables & Figures
11. Annexure.

The printed text of dissertation should not be less than 50 pages/2500 words and shall not exceed 75 pages excluding references, tables, questionnaires and other annexure. It should be neatly typed in double line spacing (Font 12, Times New Roman) on one side of paper (A4 Size, 8.27" X 11.69") and Hard bound properly (No Spiral binding). Four copies of dissertation thus prepared shall be submitted to the Registrar (Evaluation), one month before final examination on or before the dates notified by the university duly certified by the guide, head of the department and head of the institution. In the Dissertation the Candidate should not disclose his Identity or of the Guide or Institution in anyway.

Three examiners appointed by the university shall evaluate the dissertation. Two of the examiners (external) shall be from outside university and one examiner (internal) will be from the institute.

A candidate who has submitted his/her dissertation once is not required to submit a fresh dissertation if he/she reappears for the examination in the same branch on the subsequent occasion, provided the dissertation has been accepted by the examiners.

If the student has submitted her examination form & also his/her dissertation previously,

he/she will be permitted to give the examination within a period of 4 years anytime in future provided the dissertation has been accepted. The terms satisfactorily kept by him will be valid for a period of 4 years subsequent to submission of the dissertation after which he/she will have to undergo Post-graduate training again for terms to be eligible for appearing for theory & Practical examination.

POST-GRADUATE GUIDE:

A PG guide must have a Post-Graduate Degree in Physiotherapy with at-least 5 years of full timeteaching in the core areas of physiotherapy after post-graduation. Notwithstanding the above clause, in a case of acute shortage of qualified Post-Graduate guides, a PG teacher with 3 years full time teaching experience after Masters Degree can be considered. The PG guide can only guide student belonging to his/her specialized branch. These clauses may be reviewed after two years. The age of teacher /guide shall not exceed 62 years and the guide student ratio shall be 1: 3.

Co-guide: may be included provided the work requires substantial contribution from a sister departmentor from another medical institution recognized for teaching /training by the University. The co- guide shall be a recognized postgraduate teacher of the University

Change of Guide: In the event of a recognized guide leaving the college for any reason or in the event of death of guide, another recognized guide may take over the duties of the guide with prior permission from the university.

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Identify, analyse and interpret suitable data to enable the research question to be answered
CO2	Demonstrate the ability to collate and critically assess/interpret data
CO3	Develop an ability to effectively communicate knowledge in a scientific manner
CO4	Describe the process of carrying out independent research in written format and report your results and conclusions with reference to existing literature
CO5	Work autonomously in an effective manner, setting and meeting deadlines
CO6	Present the research effectively in a conference setting or a written publication

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	-	2	2	2	-	1
CO2	1	-	3	1	-	1	1
CO3	3	1	2	-	-	-	1
CO4	1	-	3	-	-	1	-
CO5	-	-	1	1	3	2	3
CO6	3	-	-	1	1	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”